1 Abstract

Under-five mortality has fallen globally, but large inequalities remain between and within countries. Standard decomposition methods can show which variables contribute to socioeconomic inequality, but they are less direct in showing how determinants combine into subgroups. This thesis developed and applied a tree-based method for analysing socioeconomic inequality in under-five mortality. The method used the concentration index as the impurity measure in recursive partitioning, so that splits were chosen to reduce within-node socioeconomic inequality in the outcome. The method was applied to child-level data from the Democratic Republic of the Congo DHS phase V survey, using under-five mortality as the outcome and the DHS wealth index as the socioeconomic ranking variable.

The analysis compared indirect decomposition methods with direct concentration-index trees and forests. In the indirect methods, a model was fitted first, either a generalised linear model or a predictive model such as a random forest, and the fitted model was then used to decompose inequality. The regression decomposition placed the greatest weight on the mother's education and the father's agricultural occupation. The SHAP decomposition was less concentrated on these two variables. Residence, household wealth, mother's education, skilled birth attendance, and parental occupation all appeared across the criteria.

The direct trees added the subgroup part of the analysis. The rank-dependent trees based on standard concentration index(CI), generalised concentration index(CIg), and corrected concentration index(CIc) split mainly through mother's education, father's agricultural occupation, skilled birth attendance, household wealth, and Katanga. The high-mortality branch was made up of children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga. This pattern was clear in the fitted data, but it did not carry well to the validation folds. The level-dependent concentration index(L) method gave a different tree. It was first split by residence, then by occupation, wealth, and province. The L tree and forest both kept positive validation gains, so the level-dependent result was treated as the more stable direct result in this application.

The thesis shows that concentration-index trees can be used to add subgroup rules to the decomposition of health inequality. The fitted tree can be read as a decision rule, making the result easier to discuss than a table of percentage contributions alone.

2 Introduction

Child survival has improved since 1990, but the gains have not closed the gap between countries or between groups of children. Globally, under-five mortality fell from 93 deaths per 1,000 live births in 1990 to 37.7 in 2019, and the number of under-five deaths fell from 12.5 million to 5.2 million (Sharrow *et al.*, 2022). Reduction in under-5 mortality are linked to improved child health services, greater access to care, and lower levels of extreme poverty (Jones *et al.*, 2003). The burden, however, is still concentrated in poorer settings. The latest (World Health Organisation, 2026) estimates show that sub-Saharan Africa had the highest regional under-five mortality rate, 67 per 1000 live births, and almost half of all under-five deaths occurred in five countries, including the Democratic Republic of the Congo (Sharrow *et al.*, 2022). Within these countries, a child's risk also depends on household poverty, proximity to health services, access to clean water, sanitation, and health information, and place of residence. Even a better-off household may remain at risk when local services are weak or far away. For this reason, studying socioeconomic inequality in under-five mortality helps show which children are still exposed to risks that can be prevented.

When statistical methods such as regression analysis are used in policy making, a national estimate may show that poorer children have higher mortality rates. However, it can hide subgroup differences linked to household conditions, province(administrative areas), maternal background, or demographic characteristics. Identifying risk groups matters because interventions are organised in real settings, through health facilities, communities, provinces(administrative areas), and programme catchment areas. A more detailed inequality analysis can therefore identify the groups that should receive attention and indicate where a single public health response may not be sufficient.

Health inequality analysis commonly begins by ordering individuals or households according to socioeconomic position and then examining how the health outcome is distributed along that ordering. The concentration index provides a standard way to summarise this relationship (Wagstaff, van Doorslaer and Watanabe, 2003). Classical decomposition methods then use a regression model to estimate how different determinants contribute to the observed inequality in the outcome (Wagstaff, van Doorslaer and Watanabe, 2003; Speybroeck *et al.*, 2010). This is useful because it gives a direct additive breakdown. At the same time, the interpretation depends on the average effects from the fitted model. An important structure can therefore be missed if mortality risk varies across subgroups, if determinants act jointly, or if the relationship differs above and below certain thresholds.

In this thesis, we develop a tree-based way of decomposing socioeconomic inequality in health. The method still uses the concentration index, but places it inside the splitting process of a tree. A typical decision tree chooses a split because it improves predictive performance. Here, the split is chosen because it separates the data into groups where socioeconomic inequality in the outcome is lower within each group. The fitted tree then produces decision rules that describe population groups. These rules are easier to discuss in public health work than regression coefficients alone, because they point to groups that policy makers, programme teams, and public health planners can recognise and act on.

At each node, the tree searches across candidate variables and admissible split points, then selects the variable and split pair that gives the largest reduction in weighted within-node concentration-index impurity. The fitted tree is stored using the partykit tree representation for plotting and traversal (Hothorn and Zeileis, 2015), but the tree-growing rule is not the conditional-inference variable-selection rule. We then apply the same split logic in a random forest setting. The forest is useful because it averages many inequality-oriented trees and is less dependent on any single early split, although this also makes the fitted structure less transparent. To recover a readable summary of the forest, we fit a surrogate tree to the forest predictions. This follows the idea of using a representative tree to summarise the main behaviour of an ensemble in a form that can be read as decision rules (Di Castro and Bertini, 2019). Finally, we use SHAP values to express the fitted tree-based predictions as additive feature contributions (Lundberg and Lee, 2017; Lundberg, Erion and Lee, 2018). The surrogate tree summarises the main subgroup structure, while SHAP provides contributions at the level of the determinants.

For the analysis, we use child-level data from the Democratic Republic of the Congo DHS phase V survey (The DRC Program Program, 2007). Under-five mortality is treated as the outcome, and the DHS household wealth index serves as the socioeconomic ranking variable.

This is appropriate for the DRC setting, where DHS-based research has shown that poverty and education are associated with access to health insurance (Tsala Dimbuene, Muanza Nzuzi and Nzita Kikhela, 2022). The splitting variables cover characteristics of the child, the mother, the household, occupation, residence, and province.

The thesis is organised around four objectives. The first is to define the concentration index and demonstrate how it can serve as a splitting criterion in recursive partitioning. The second is to apply concentration-index trees and forests to under-five mortality using the DRC DHS data. The third is to examine whether different concentration-based impurity measures yield distinct split choices. The fourth is to use SHAP-based contributions to decompose the concentration index of the fitted tree-based predictions.

3 Methodology

3.1 Data, Outcome, and Determinants

This thesis uses child-level data from the Democratic Republic of the Congo DHS Phase V survey (The DRC Program Program, 2007). The analysis includes children with complete information on the outcome, household wealth score, survey weight, and covariates used in the tree models. DHS sampling weights are rescaled and used as non-negative case weights in the inequality-tree fitting procedure.

The outcome is under-five mortality, a binary adverse outcome:

$$Y_i = \begin{cases} 1, & \text{if child } i \text{ died before 60 months of age,} \\ 0, & \text{otherwise.} \end{cases}$$

The coding uses the child's survival status and age at death. Children who died before 60 months receive a value of one. Since the outcome is adverse, higher fitted values correspond to higher mortality risk.

Socioeconomic position is measured using the DHS household wealth index(The DHS Program, 2007). The wealth score is a PCA-derived composite measure of household living standard based on asset ownership, housing materials, water access, sanitation facilities, and related household amenities. The continuous wealth score is used to rank children from poorer to richer households when computing the concentration index. Wealth defines the socioeconomic scale along which mortality inequality is measured.

The independent variables describe the child, mother, household, occupation, residence, and province. They include child sex, birth-order and birth-interval group, mother's age group, rural residence, mother's and partner's education, mother's and partner's occupation, whether either parent belongs to a low-skill or non-working occupational group, and province. Categorical variables are recoded into interpretable groups before the tree is fitted.

3.2 Measuring Socioeconomic Inequality in Health

This methodology aims to develop a decomposition of health inequalities using tree methods. We will first introduce the concepts of health inequality decomposition, then move to tree-based methods.

3.2.1 The Concentration Index

The concentration index measures socioeconomic-related inequality in a health variable. It uses the concentration curve, which plots the cumulative proportion of the health variable on the vertical axis against the cumulative proportion of the population on the horizontal axis. Individuals are ordered from the most disadvantaged to the most advantaged before constructing the curve (Kakwani, Wagstaff and van Doorslaer, 1997; Wagstaff, van Doorslaer and Watanabe, 2003). When the curve follows the 45-degree line, the health variable has the same distribution across the socioeconomic ranking.

y_i denotes the health variable for individual i, and let R_i denote that individual's fractional socioeconomic rank, ordered from poorest to richest. For an unweighted sample, the fractional rank is

$$R_i = \frac{i - 0.5}{n},$$

where $i = 1, \ldots, n$ denotes the individual's position in the socioeconomic ranking. If μ is the mean of y, the concentration index is

$$C = \frac{2}{n\mu} \sum_{i=1}^{n} y_i R_i - 1.$$

A value of zero indicates no socioeconomic-related inequality in the health variable. A negative value indicates that the variable is concentrated among poorer individuals, while a positive value indicates that it is concentrated among richer individuals. The sign, therefore, depends on the meaning of the health variable. For adverse outcomes such as mortality, illness, or malnutrition, a negative concentration index indicates inequality that disadvantages people with low incomes.

In the simulated example in Figure 1, we simulate the health outcome to be concentrated among poorer individuals, so the curve lies above the 45-degree line and the concentration index is negative (-0.314).

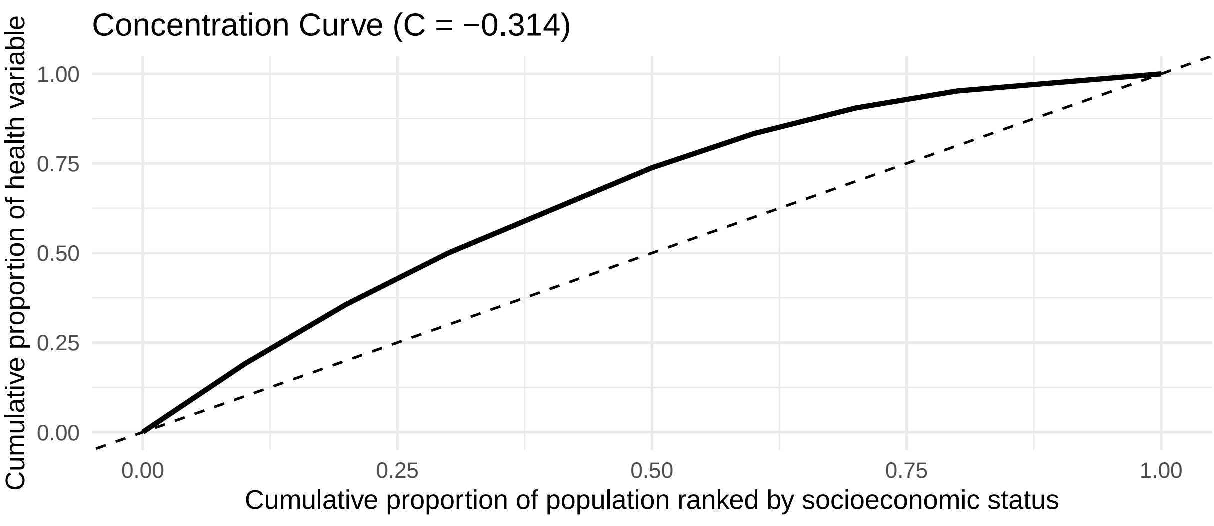

Figure 1: Illustration of a concentration curve using a simple hypothetical example.

3.2.2 Decomposing Socioeconomic Inequality in Health

Inequality is measured in the observed health outcome y_i, but interventions usually work through the factors that affect health rather than through the outcome itself. Decomposition helps relate inequality in y_i to the observed social determinants that may explain it.

Let the health outcome for the individual i be defined.

$$y_i = f(x_i) + \varepsilon_i,$$

where x_i denotes the observed determinants and ε_i represents the part that the model does not explain. Let $\hat{y}_i = f(x_i)$ be the fitted value, and use this fitted model to decompose the concentration index.

$$C = \frac{2}{n\mu} \sum_{i=1}^{n} \hat{y}_i R_i - 1,$$

where:

$\hat{y}_i$ is the estimated health outcome for an individual i based on the observed determinants x_i. (Wagstaff, van Doorslaer and Watanabe, 2003) showed that the decomposition of the concentration index results in the following expressions:

$$C = \sum_k \left(\frac{\beta_k \bar{x}_k}{\mu} \right) C_k + \frac{GC_\varepsilon}{\mu}.$$

In this expression, C is the concentration index of the health outcome, μ is the mean of that outcome, β_k is the regression coefficient for determinant x_k, $\bar{x}_k$ is the mean of x_k, and C_k is the concentration index of that determinant. The last term, GC_ε, is the generalised concentration index of the regression residuals. From this, we can infer that the contribution to the determinant x_k is

$$\text{Contribution of } x_k = \left(\frac{\beta_k \bar{x}_k}{\mu} \right) C_k.$$

The concentration index(C_k) for determinant x_k can be written as follows

$$C_k = \frac{2}{n\bar{x}_k} \sum_{i=1}^{n} x_{ki} R_i - 1.$$

where x_{ki} is the value of determinant x_k for the individual i, $\bar{x}_k$ is the mean of determinant x_k, and R_i is the fractional socioeconomic rank of individual i.

3.2.3 Percentage Contributions

In the concentration index results, the percentage contribution for each determinant is usually reported. For determinant k, the total contribution is

$$\text{AC}_k = \left(\frac{\beta_k \bar{x}_k}{\mu} \right) C_k.$$

The percentage contribution is

$$\mathrm{PC}_k = 100 \times \frac{\mathrm{AC}_k}{C}.$$

This is a signed share of total socioeconomic inequality. In our analysis of under-5 mortality, a positive value means that the determinant contributes in the same direction as the observed mortality inequality (reinforces the observed pattern). A negative value means that it offsets part of that inequality. A percentage can exceed 100 per cent when other determinants, or the residual term, contribute in the opposite direction.

3.2.4 Concentration Index Variants and a Level-Dependent Alternative

When the outcome is bounded, for instance, in our case, a binary outcome, (Wagstaff, 2005) showed that the lower and upper bounds depend on the outcome's mean, so the feasible range narrows as the mean increases. That means a change in the index can reflect not only changes in socioeconomic inequality but also changes in the average level of the outcome. (Erreygers, 2009) shows that this issue goes beyond the binary case. In general, once a health variable has a finite upper bound or a positive lower bound, the bounds of the concentration index vary with the mean. For this reason, several related measures have been introduced, rather than relying solely on the standard index. In this thesis, the rank-dependent measures are the standard concentration index (CI), the generalised concentration index (CIg), and the corrected concentration index for bounded outcomes (CIc). A level-dependent alternative, denoted L, is also considered, which is important when a socioeconomic variable has a defensible level interpretation (Erreygers and Kessels, 2017), for example, income.

The standard concentration index is

$$CI_y = \frac{2}{\mu_y} \operatorname{cov}_w(y_i, R_i),$$

where y_i is the health outcome, R_i is the fractional socioeconomic rank, and μ_y is the mean of the outcome.

The generalised concentration index is

$$CIg_y = 2 \operatorname{cov}_w(y_i, R_i) = \mu_y CI_y.$$

For bounded outcomes, the corrected concentration index takes the form.

$$CIc_y = \frac{4}{b-a} CIg_y = \frac{8}{b-a} \operatorname{cov}_w(y_i, R_i),$$

where a and b are the lower and upper bounds of the outcome. For a binary outcome, where $a = 0$ and $b = 1$, this reduces to

$$CIc_y = 4\mu_y CI_y.$$

The level-dependent index L uses observed socioeconomic levels directly rather than converting them into ranks:

$$L_y = \sum_{i=1}^{n} p_i \left(\frac{s_i - \mu_s}{\mu_s} \right) y_i, \qquad p_i = \frac{w_i}{\sum_{j=1}^{n} w_j}, \qquad \mu_s = \sum_{i=1}^{n} p_i s_i.$$

s_i is the observed socioeconomic level, y_i is the health outcome, and w_i is the case weight. Unlike CI, CIg, and CIc, the L index does not replace s_i with a fractional rank.

The DHS wealth variable used in this thesis is the continuous wealth-index factor score, $v191$. It is not observed income, consumption, or expenditure, but a PCA-derived composite measure of household living standard based on asset ownership, housing materials, water access, sanitation facilities, and related household amenities (The DHS Program, 2007). For the L index (Erreygers and Kessels, 2017), state that the level-dependent construction is acceptable only when the socioeconomic variable has ratio-scale properties; since the wealth index is a PCA score rather than income or consumption, results based on L should therefore be interpreted with caution. When the concentration indeces are used as impurity measures, the tree may not form the same subgroups under all concentration indices. A split chosen under CI may differ from one chosen under CIg, CIc, or L, because each index defines inequality in a slightly different way.

3.2.5 Limits of the Linear Framework

The standard decomposition uses a generalised linear model, $y_i = \beta_0 + \sum_k \beta_k x_{ki} + \varepsilon_i$, where β_0 is the intercept, β_k is the coefficient associated with determinant x_k, x_{ki} is the value of determinant k for individual i, and ε_i is the error term.

In the regression-based decomposition framework, each determinant contributes through a fitted regression coefficient (Wagstaff, van Doorslaer and Watanabe, 2003; Speybroeck *et al.*, 2010). This makes the decomposition clear, because the contribution of each variable can be written as part of an additive summary. The limitation is that the coefficient is an estimated conditional average effect over the population. The effect of a determinant may differ by household conditions, province, maternal background, or access to services. Some effects may also appear only after a threshold is crossed. This is why this thesis next turns to tree-based methods, which can represent thresholds and interactions more directly without specifying this before fitting a model.

3.3 Tree-Based Models for Inequality Analysis

3.3.1 Recursive Partitioning and Decision Trees

A decision tree does not fit a single global equation to all observations per determinant. It divides the data into subgroups using a sequence of rules based on the predictors (Hastie, Tibshirani and Friedman, 2017). Each path through the tree leads to a terminal node, or leaf, which represents a subgroup of observations with similar values on the variables used for splitting.

In a regression tree, the fitted value for a leaf is the mean outcome in that subgroup. In a classification tree, it is usually a class label or an estimated class probability. The model is therefore piecewise: it gives different summaries for different parts of the covariate space. If the terminal regions are denoted by $R_1, \dots, R_M$, the fitted regression tree can be written as

$$\hat{f}(x) = \sum_{m=1}^{M} c_m I(x \in R_m),$$

where c_m is the fitted value assigned to region R_m and $I(x \in R_m)$ equals 1 if observation x falls in that region and 0 otherwise. Below Figure 2 is an illustration drawn from the Democratic Republic of the Congo (DRC) household survey data. We fit a classification tree to predict the probability of under-five mortality using two continuous predictors: the child's age in months and the household wealth index. The left panel shows the tree itself. The right panel shows the same fitted tree as a partition of the two-dimensional predictor space. Each rectangle corresponds to one terminal node, and all observations inside the same rectangle receive the same fitted probability of mortality.

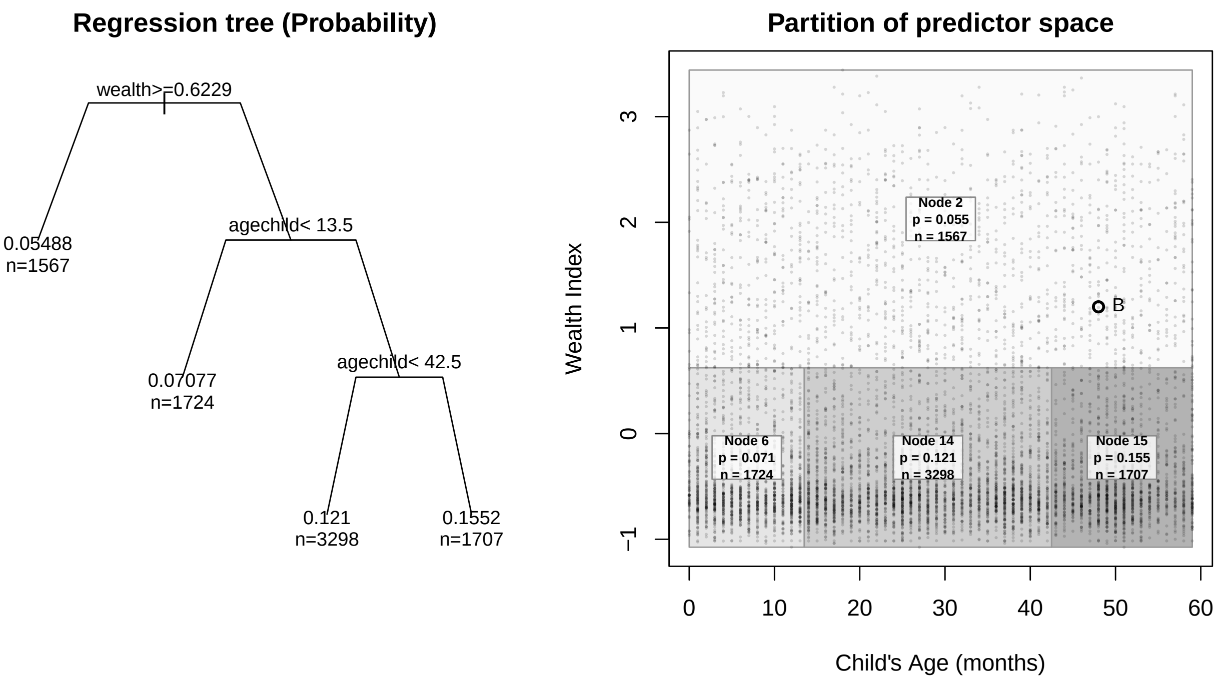

Figure 2: A classification tree predicting under-five mortality and its rectangular partition of (Age, Wealth)

From the example above Figure 2, we can see the main difference from the regression decomposition. A linear model assigns each determinant a single coefficient, and we must specify interactions beforehand. The same conditional average effect(coefficient) is used for the whole sample. A tree first splits the data into subgroups and then assigns fitted values to each subgroup. The relationship between the outcome and the determinants can therefore change from one part of the population to another. A decision tree thus becomes useful when the pattern depends on thresholds, interactions, or subgroup-specific effects.

3.3.2 Recursive Partitioning for Inequality Analysis

In a usual CART(Classification and Regression Trees)(Breiman *et al.*, 1984), splits are chosen mainly for prediction. A regression tree searches for splits that reduce within-node outcome variation, while a classification tree searches for splits that improve class separation (Hastie, Tibshirani and Friedman, 2017).

In this thesis, the split target is changed. The tree is used to form subgroups where under-five mortality is less concentrated by socioeconomic position. The outcome level in each subgroup is still reported, but it is not the quantity used to choose the split. A split is preferred when it reduces the within-node concentration index of under-five mortality.

3.3.3 Concentration-Index-Based Recursive Partitioning

To implement recursive partitioning using concentration indices, we developed a greedy concentration-index tree. The procedure follows the CART method of recursive partitioning but replaces the prediction impurity with a concentration-index impurity. The gain function selects both the variable and the split point. At node t, the selected split is

$$(j^*, s^*) = \arg \max_{j \in \mathcal{J}_t,\, s \in \mathcal{S}_{jt}} G_m(j, s, t),$$

where m is one of CI, CIg, or CIc, $\mathcal{J}_t$ is the set of candidate variables at node t, and $\mathcal{S}_{jt}$ is the set of admissible splits for variable j. The implementation uses partykit tree representation (Hothorn and Zeileis, 2015). However, it replaces the conditional-inference tree-growing engine with a custom recursive builder whose split rule directly maximises concentration-index gain over candidate variables and split points.

Weighted Within-Node Rank Function

The concentration index requires individuals to be ranked from poorest to richest. Because the tree is recursive, this ranking must be recomputed at each node. We define a weighted within-node rank function. For individual i belonging to node t, let the weighted fractional rank be

$$R_{i|t} = \frac{1}{W_t} \left(\sum_{j \in t : S_j < S_i} w_j + \frac{w_i}{2} \right),$$

where S_i denotes the socioeconomic ranking value of individual i, and S_j denotes the corresponding value for individual j in the same node. The terms w_i and w_j are their survey weights, and $W_t = \sum_{i \in t} w_i$ is the total survey weight in node t.

In this formula, the first term counts the survey weight of children in the same node who are poorer than child i, and adding half of w_i places the child at the midpoint of its own weighted position in the ordered list. The rank is then divided by the total node weight, so $R_{i|t}$ lies between 0 and 1. Since this calculation is repeated separately inside each node, the wealth rank used by the concentration index is local to the subgroup being split, not to the full DHS sample.

Node Impurity Based on the Concentration Index

After we define within-node rank, the tree needs a criterion for comparing possible splits. In ordinary CART, node impurity is based on the homogeneity of the outcomes. A regression tree favours splits that reduce within-node variance or residual sum of squares, while a classification tree may use classification error, Gini impurity, or entropy (Hastie, Tibshirani and Friedman, 2017). A good split should leave the two child nodes less mixed in the outcome than the parent node. In this thesis, the same node impurity logic is used, but the impurity measure is replaced by one based on the concentration index.

In the present method, the aim is not to make nodes homogeneous in their outcome level, but to make them more homogeneous with respect to socioeconomic inequality in the outcome. We therefore define the impurity of a node as the absolute concentration index within that node. For node t, let

$$CI_t = \frac{2}{W_t \mu_t} \sum_{i \in t} w_i Y_i R_{i|t} - 1$$

where

$$\mu_t = \frac{1}{W_t} \sum_{i \in t} w_i Y_i$$

is the weighted mean of the health outcome in node t. This may be written in weighted covariance form as

$$CI_t = \frac{2}{\mu_t} \operatorname{cov}_w(Y_i, R_{i|t}).$$

The quantity CI_t measures whether the health outcome is concentrated among poorer or richer individuals within the node. Values close to zero indicate little socioeconomic inequality within the subgroup, while larger absolute values indicate stronger inequality. Because the split criterion is intended to measure the strength of inequality remaining within a node, regardless of whether it is pro-poor or pro-rich, we use the absolute value as the impurity measure:

$$I(t) = |CI_t|.$$

Split Gain Function

At this stage, the tree needs an objective function: a rule for deciding which split is best. At each node, the objective searches for candidate splits and keeps the one that yields the largest improvement under the chosen criterion.

In this thesis, the same search over candidate splits used in classification and regression trees (CART)is retained, but the objective function is changed. The split is not selected because it gives the best prediction of under-five mortality in the usual trees. It is selected because it gives the largest reduction in socioeconomic inequality in under-five mortality.

We therefore define node impurity as the absolute value of the chosen concentration-index measure within that node. A split is preferred when the two child nodes contain less within-node socioeconomic inequality in mortality than the parent node. For a candidate variable-split pair (j, s) that divides node t into a left child node $t_L(j, s)$ and a right child node $t_R(j, s)$, the weighted reduction in impurity is defined as

$$G_m(j, s, t) = I_m(t) - \left(\frac{W_{t_L(j,s)}}{W_t} I_m(t_L(j, s)) + \frac{W_{t_R(j,s)}}{W_t} I_m(t_R(j, s)) \right).$$

$I_m(t)$ is the concentration-index impurity in the parent node, while $I_m(t_L(j, s))$ and $I_m(t_R(j, s))$ are the impurities in the two child nodes created by split (j, s).

In this implementation, the gain from a split is the reduction in weighted absolute concentration-index impurity from the parent node to the two child nodes. If a split sends observations into two groups where the weighted child concentration indices are closer to zero than the parent value, the split receives a positive gain. The tree, therefore, grows by selecting variables that reduce within-node socioeconomic inequality, rather than by selecting variables that minimise prediction error. After the split, the resulting child nodes can be compared by their outcome levels and remaining inequality. Since $I_m(t)$ is fixed for a given parent node, maximising $G_m(j, s, t)$ is the same as choosing the variable-split pair that leaves the least weighted inequality in the child nodes. The selected split is

$$(j^*, s^*) = \arg \max_{j \in \mathcal{J}_t,\, s \in \mathcal{S}_{jt}} G_m(j, s, t).$$

The selected split is the candidate pair (j, s) with the largest admissible gain.

Split behaviour under alternative concentration-based impurity measures

The impurity measure is not separate from the gain function. It enters directly into the definition of the gain. Once an impurity measure has been chosen, the gain function uses that measure to compare and rank all admissible candidate splits at the current node.

For a candidate split (j, s) at node t, the gain under impurity measure m is

$$G_m(j, s, t) = I_m(t) - \left[p_L I_m(t_L) + p_R I_m(t_R)\right],$$

where $I_m(t)$ is the impurity of the parent node, and $I_m(t_L)$ and $I_m(t_R)$ are the impurities of the left and right child nodes, the terms p_L and p_R weight the child-node impurities by their survey-weighted node sizes.

At a fixed parent node, $I_m(t)$ is the same for all candidate splits evaluated under the same impurity measure. Therefore, maximising the gain is equivalent to minimising the weighted impurity left in the child nodes:

$$p_L I_m(t_L) + p_R I_m(t_R).$$

Thus, the impurity measure affects the gain function, which in turn ranks candidate splits. The concentration indices here are impurity measures that are intended to be reduced. Under the standard concentration index rule, the child-node impurity is relative to the child-node mean:

$$p_L \frac{A_{t_L}}{\mu_{t_L}} + p_R \frac{A_{t_R}}{\mu_{t_R}}.$$

Under the generalised concentration index rule, the criterion remains on the absolute scale:

$$p_L A_{t_L} + p_R A_{t_R}.$$

Because the child-node means μ_{t_L} and μ_{t_R} can vary across candidate splits, the CI and CIg rules may rank the same splits differently. The corrected concentration index CIc is different in scale but not in ordering when the outcome range $[a, b]$ is fixed, since

$$G_{CIc}(j,s,t) = \frac{4}{b-a} G_{CIg}(j,s,t).$$

$\frac{4}{b-a}$ is positive and independent of the candidate split. Therefore, CIc and CIg may produce the same split ranking, although a fixed minimum-gain threshold would need to be rescaled if the stopping rule uses the raw gain value.

In short, the impurity measure determines what the gain function measures and the scale, and the gain function determines which split is selected.

Split Search for Numeric and Categorical Predictors

At each node, the algorithm searches across the candidate predictors and their admissible binary partitions. The same gain function is used for both numeric and categorical predictors, but the candidate splits are constructed differently. For a numeric predictor X_k, the values observed in node t are sorted, and possible cutpoints are placed only between distinct adjacent values. For two adjacent ordered values, $x_{(r)}$ and $x_{(r+1)}$, the candidate cutpoint is placed halfway between them:

$$s = \frac{x_{(r)} + x_{(r+1)}}{2}.$$

This gives the binary rule.

$$X_k \leq s \qquad \text{versus} \qquad X_k > s.$$

Only cutpoints that leave enough survey weight in both child nodes are retained. For each numeric predictor, the best admissible cutpoint is the one with the largest concentration-index gain:

$$s_k^* = \arg\max_s G_m(k,s,t).$$

For a categorical predictor X_k, the levels present in node t are first ordered by their weighted mean outcome. If these ordered levels are $l_1, \dots, l_J$, the candidate splits are formed cumulatively:

$$\{l_1\} \quad \text{versus} \quad \{l_2, \dots, l_J\},$$

$$\{l_1, l_2\} \quad \text{versus} \quad \{l_3, \dots, l_J\},$$

up to

$$\{l_1, \dots, l_{J-1}\} \quad \text{versus} \quad \{l_J\}.$$

The eligible splits are subject to the stopping rules discussed in the next section.

Stopping Rules

A node is split only if it passes the node level checks, has at least one admissible candidate split, and the best candidate gives enough improvement. First, node t is searched only when its depth is below the maximum tree depth (maxdepth), and its weighted sample size is at least the minimum weighted parent-node size (minsplit):

$$d_t < d_{\max} \quad \text{and} \quad W_t \geq W_{\text{minsplit}}.$$

If either condition fails, the node is terminal. For a node that passes maxdepth and minsplit checks, each candidate split (j, s) must create two child nodes that are large enough. A split is admissible only if both children satisfy the minimum weighted child-node size (minbucket),

$$W_{t_L(j,s)} \geq W_{\text{minbucket}}, \quad W_{t_R(j,s)} \geq W_{\text{minbucket}},$$

and the minimum child-node weight proportion (minprob),

$$\frac{W_{t_L(j,s)}}{W_t} \geq p_{\min}, \quad \frac{W_{t_R(j,s)}}{W_t} \geq p_{\min}.$$

These two rules remove unsuitable candidate splits before the gain is compared. A node becomes terminal if no candidate split remains after this filtering. Among the admissible splits, the selected split must have a finite gain and must exceed the minimum absolute gain (min_gain):

$$G_m(j^*, s^*, t) > G_{\min}.$$

The algorithm also checks the minimum relative gain (min_relative_gain). For split (j, s), this is defined as

$$RG_m(j, s, t) = \frac{G_m(j, s, t)}{|I_m(t)|}.$$

This expresses the gain as a share of the impurity in the parent node. It helps avoid splits that have a positive raw gain but reduce only a very small part of the parent-node impurity. It also makes the stopping rule less dependent on the numerical scale of the chosen concentration-index impurity measure. Thus, a node becomes terminal when the node-level checks fail, when no split satisfies the child-size rules, or when the best admissible split fails the absolute or relative gain requirement. The number of predictors considered (mtry) only changes which predictors are searched at a node, especially in the forest setting. It is not a stopping rule. The full algorithm is shown in pseudocode and flowchart form in the appendix (Figure 14), and the implementation is available in the ineqTrees package (Mburu, 2026).

3.3.4 Extensions to Ensemble Methods: Random Forests

The main advantage of the concentration-index tree is that the subgroup structure can be inspected directly. Observed covariates define each branch, and each terminal node can be described by its weighted sample size, mean outcome, and remaining concentration-index value.

A limitation of the single tree is that its structure can be unstable. The algorithm grows the tree greedily, so an early split determines which observations are available for later splits. A small change in the data or in the tuning controls can therefore change the upper part of the tree and lead to a different set of terminal nodes. Tree depth also matters. A shallow tree may miss important combinations of determinants, while a deeper tree may start to follow small weighted subgroups rather than a pattern that is likely to appear again in another sample.

To reduce this dependence on one fitted tree, we extend the concentration-index split rule to a random forest. The forest fits many greedy CI trees to perturbed versions of the training data. Within each tree, node splitting follows the same direct maximisation of concentration-index gain over admissible variable-split pairs used by the single-concentration-index tree. Randomness enters through row perturbation and, optionally, through mtry, the number of candidate predictors searched at each node. The final prediction is obtained by aggregating predictions across trees.

The forest contains B trees, and $\hat{f}_b(x)$ is the prediction from tree b, the forest prediction is

$$\hat{f}_{RF}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}_b(x).$$

The interpretation of $\hat{f}_{RF}(x)$ follows from how the forest is fitted. Since the trees are grown to reduce concentration-index impurity, not to minimise prediction error, we do not read the forest output as a fully calibrated probability of under-five death. It is a mortality-scale score obtained by averaging terminal-node mortality values across many inequality-based trees. We use it to describe the subgroup structure learned by the forest, not as the main target of a predictive risk model.

Averaging over many such inequality-oriented trees makes the fitted values less dependent on any single partition. It allows the model to reflect more complex combinations of child, maternal, household, and geographic characteristics. The cost is that the final forest is no longer explainable. To address unexplainability, we fit a surrogate tree (Di Castro and Bertini, 2019) to summarise the forest. The surrogate tree uses the forest prediction $\hat{f}_{RF}(x_i)$ as its outcome, with the same candidate predictors used in the main analysis. It does not replace the forest. Its role is to give a smaller set of rules that approximates the main subgroup structure learned by the forest. The terminal nodes of this surrogate tree can then be described using their average forest-predicted mortality risk and their concentration index. This gives a readable summary of how the forest separates children into inequality-relevant groups, while keeping the random forest as the more flexible model.

3.3.5 Evaluation and Model Selection of Tree-Based Methods for Inequality Analysis

The purpose of the inequality tree is to split the sample into subgroups where under-five mortality is no longer strongly concentrated by socioeconomic position. In an ideal terminal node, the selected inequality index would be close to zero. Model selection, therefore, evaluates how much of the root-node impurity is removed after partitioning, and how much impurity remains inside the terminal nodes. For a fitted tree, the remaining impurity is summarised by the weighted terminal-node impurity:

$$\sum_{\ell \in \mathcal{L}} \frac{W_\ell}{W} I_m(D_\ell).$$

Here $\mathcal{L}$ is the set of terminal nodes, W_ℓ/W is the share of total survey weight in terminal node ℓ, and $I_m(D_\ell)$ is the selected node impurity. For CI, CIg, and CIc, $I_m(D\ell)$ is the absolute value of the selected concentration index. For the level-dependent measure, $I_L(D_\ell) = |L(D_\ell)|$. Lower values mean that the terminal nodes are closer to the target of zero within-node socioeconomic inequality.

The stopping rules introduced earlier are treated as hyperparameters. Cross-validation is used to select settings that yield a high gain score. For validation, we use the held-out inequality gain:

$$G_m^{val}(c, v) = I_m(D_v) - \sum_{\ell \in \mathcal{L}_c} \frac{W_{\ell v}}{W_v} I_m(D_{\ell v}).$$

$I_m(D_v)$ is the impurity of the validation fold before splitting. The summation is the impurity remaining after the validation observations are dropped from the fitted tree. If $G_m^{val}(c, v)$ is close to absolute root inequality, the tree has reduced the validity on-fold impurity compared with the root node. If it is close to zero, the split structure has added a little. If it is negative, the tree has not transferred well to the held-out fold. We also scale this gain by the validation-fold root impurity:

$$RG_m^{val}(c, v) = \frac{G_m^{val}(c, v)}{I_m(D_v)}.$$

where c is the tunning setting being evaluated, v is the validation fold, and $I_m(D_v)$ is the impurity of the validation fold before splitting. This gives the proportion of root impurity removed by the fitted partition. For each impurity criterion, we choose the tuning setting with the highest average relative validation gain:

$$\overline{RG}_m^{val}(c) = \frac{1}{V} \sum_{v=1}^{V} RG_m^{val}(c, v),$$

$$c^* = \arg\max_{c \in \mathcal{C}} \overline{RG}_m^{val}(c).$$

Raw validation shows the reduction in the original scale of the chosen impurity measure. Training gain metrics are useful to judge overfitting. That is when there is a large gap between the metric chosen on the training set and that on the validation set.

3.4 SHAP based Decomposition Framework

Tree-based models, including random forests, can provide flexible predictions of health risk. This makes it hard to use them for decomposition, since the current decomposition methods rely on GLMs(Wagstaff, van Doorslaer and Watanabe, 2003; Speybroeck *et al.*, 2010). To deal with this, this thesis uses SHAP values to rewrite the fitted predictions as a sum of feature-specific contributions.

3.4.1 General Notation and Setup

We start with a dataset containing n individuals, indexed by $i = 1, \dots, n$. For each individual, we observe:

- y_i, the health outcome
- $x_i = (x_{1i}, x_{2i}, \dots, x_{Ki})$, a vector of K determinants
- SES_i, a measure of socioeconomic status
- R_i, the fractional socioeconomic rank, defined from the most disadvantaged to the most advantaged using SES_i

Let $\hat{f}(x)$ denote a fitted tree-based model, such as a random forest. The fitted outcome for individual i is

$$\hat{y}_i = \hat{f}(x_i).$$

The aim is to decompose the concentration index of the fitted outcomes. Let $\mu_{\hat{y}}$ be the sample mean of $\hat{y}_i$. Then the concentration index of the fitted values is

$$C_{\hat{y}} = \frac{2}{n\mu_{\hat{y}}} \sum_{i=1}^{n} \hat{y}_i R_i - 1.$$

where $C_{\hat{y}}$ is the concentration index of the fitted outcomes, n is the sample size, and R_i is the fractional socioeconomic rank of individual i. To obtain a decomposition, we need to express $\hat{y}_i$ as a sum of feature-specific components.

3.4.2 Tree-Based Predictions

In the regression based decomposition, the fitted value from the regression model can be separated into terms linked to the individual determinants. This makes the decomposition step relatively direct. The tree models do not have that form. A child's fitted mortality risk is determined by the path it follows through the tree and by the terminal node it reaches. In the random forest, the fitted value is an average over many such paths.

3.4.3 SHAP as the Additive Link

The tree and forest give fitted mortality rates, but those values are not estimated using determinant-level coefficients. We use SHAP values for this step. SHAP rewrites a model prediction as a baseline value plus feature contributions (Lundberg and Lee, 2017; Lundberg, Erion and Lee, 2018). For tree ensembles, TreeSHAP provides an efficient way to compute these contributions.

For child i, we write the fitted value as:

$$\hat{y}_i = \phi_0 + \sum_{k=1}^{K} \phi_{ik},$$

where ϕ_0 is the baseline fitted value and ϕ_{ik} is the SHAP contribution of determinant k for child i. In this thesis, ϕ_0 is set to the sample mean of the fitted values, $\mu_{\hat{y}}$. This gives the additive form needed for the next step: the tree-based fitted value can be inserted into the concentration-index decomposition through determinant-level SHAP contributions.

3.4.4 A SHAP-Based Definition of Decomposition

Since we have now expressed the predictions using SHAP, we can substitute this representation into the concentration index formula for the predicted outcome. We start with the fractional rank R_i and the mean prediction $\mu_{\hat{y}} = \frac{1}{n}\sum_{i=1}^{n}\hat{y}_i$, the sample concentration index of the predictions is given by:

$$C_{\hat{y}} = \frac{2}{n\mu_{\hat{y}}}\sum_{i=1}^{n}\hat{y}_i R_i - 1.$$

Recognising the property of the standard fractional rank that

$$\sum_{i=1}^{n}\left(R_i - \frac{1}{2}\right) = 0,$$

We can rewrite the concentration index formula using the centred rank $(R_i - \frac{1}{2})$. This allows us to drop the -1 term. A full proof is given in Appendix A.

$$C_{\hat{y}} = \frac{2}{n\mu_{\hat{y}}}\sum_{i=1}^{n}\hat{y}_i\left(R_i - \frac{1}{2}\right).$$

We now substitute the model-level SHAP identity $\hat{y}_i = \phi_0 + \sum_{k=1}^{K}\phi_{ik}$ into this centered formula:

$$C_{\hat{y}} = \frac{2}{n\mu_{\hat{y}}}\sum_{i=1}^{n}\left(\phi_0 + \sum_{k=1}^{K}\phi_{ik}\right)\left(R_i - \frac{1}{2}\right).$$

Expanding the expression yields:

$$C_{\hat{y}} = \frac{2}{n\mu_{\hat{y}}}\left[\phi_0\sum_{i=1}^{n}\left(R_i - \frac{1}{2}\right) + \sum_{k=1}^{K}\sum_{i=1}^{n}\phi_{ik}\left(R_i - \frac{1}{2}\right)\right].$$

Because the centred ranks sum to zero, the baseline prediction term ϕ_0 cancels out:

From this point, the concentration index of the predicted outcome simplifies to:

$$C_{\hat{y}} = \frac{2}{n\mu_{\hat{y}}}\sum_{k=1}^{K}\sum_{i=1}^{n}\phi_{ik}\left(R_i - \frac{1}{2}\right).$$

By swapping the order of summation, we isolate the deterministic components:

$$C_{\hat{y}} = \sum_{k=1}^{K}\left[\frac{2}{n\mu_{\hat{y}}}\sum_{i=1}^{n}\phi_{ik}\left(R_i - \frac{1}{2}\right)\right].$$

We define the bracketed term as the SHAP-based inequality contribution of feature k, denoted as D_k^{SHAP}:

$$D_k^{\text{SHAP}} = \frac{2}{n\mu_{\hat{y}}} \sum_{i=1}^{n} \phi_{ik} \left(R_i - \frac{1}{2} \right).$$

This results in a direct, additive SHAP decomposition for the concentration index of the predicted outcome:

$$C_{\hat{y}} = \sum_{k=1}^{K} D_k^{\text{SHAP}}.$$

This equation shows that the absolute contribution of feature k to health inequality is the rank-weighted average of its local SHAP contributions, scaled by the mean predicted outcome. A feature contributes a large share to overall inequality when its SHAP values are large in magnitude and systematically concentrated among richer or poorer individuals.

3.4.5 Percentage Contributions

As in Percentage Contributions for decomposition based on Regression Models, I express the SHAP-based contributions as percentages of the concentration index for the fitted mortality risks. For determinant k, this is

$$\text{PC}_k^{\text{SHAP}} = 100 \times \frac{D_k^{\text{SHAP}}}{C_{\hat{y}}}.$$

In this case, D_k^{SHAP} is the absolute inequality contribution from the SHAP values for determinant k, and $C_{\hat{y}}$ is the concentration index of the fitted under-five mortality risks. A positive percentage means that the determinant contributes in the same direction as the overall predicted inequality. A negative percentage means that it offsets part of that inequality. If the SHAP additivity gap is small and $C_{\hat{y}} \neq 0$, the percentages should add up to about 100 per cent.

3.4.6 Comparison with the Linear Decomposition Logic

The SHAP-based decomposition plays the same role as the Wagstaff decomposition, but it uses a different object inside the contribution term. In the regression-based framework, the absolute contribution of determinant k is

$$\left(\frac{\beta_k \bar{x}_k}{\mu} \right) C_k.$$

This combines the fitted regression coefficient with the socioeconomic distribution of the determinant. The limitation is that β_k is one conditional average association for the full sample. For example, a variable such as rural residence is treated with a single coefficient, even though its role in under-five mortality may differ across provinces, household conditions, and access to services. In the tree-based decomposition, this single coefficient is replaced by individual SHAP contributions. The absolute contribution becomes

$$D_k^{\text{SHAP}} = \frac{2}{n\mu_{\hat{y}}} \sum_{i=1}^{n} \phi_{ik} \left(R_i - \frac{1}{2} \right).$$

Here, the contribution depends on how the SHAP value for determinant k varies across the socioeconomic ranking. This allows the role of a determinant to differ across children and subgroups. If rural residence matters more in some provinces than in others, or only when combined with other household disadvantages, that pattern is carried by the individual values ϕ_{ik} rather than by a single global coefficient. The SHAP decomposition applies to the model-fitted outcome, so it decomposes $C_{\hat{y}}$ rather than C_y. The observed outcome can still be written as the fitted value plus a residual:

$$y_i = \hat{y}_i + e_i.$$

Using this identity, the concentration index of the observed outcome can be separated into the inequality explained by the fitted values and the inequality left in the residuals:

$$C_y = \frac{\mu_{\hat{y}}}{\mu_y} C_{\hat{y}} + \frac{2}{n\mu_y} \sum_{i=1}^{n} e_i \left(R_i - \frac{1}{2} \right).$$

where C_y is the concentration index of the observed outcome, $C_{\hat{y}}$ is the concentration index of the fitted values, μ_y and $\mu_{\hat{y}}$ are their means, and $e_i = y_i - \hat{y}_i$. The second term is the residual inequality not captured by the fitted tree-based model.

3.4.7 Equivalence under Linearity

Next we test whether SHAP decomposition extension safely reduces to the classical linear framework when the underlying model is structurally linear, we consider a linear regression model:

$$\hat{y}_i = \hat{\beta}_0 + \sum_{k=1}^{K} \hat{\beta}_k x_{ki}.$$

According to the mathematical properties of Shapley values, under a purely linear data-generating process, the SHAP value for a given feature k and individual i perfectly simplifies to the estimated coefficient multiplied by the mean-centred covariate (Lundberg and Lee, 2017)

$$\phi_{ik} = \hat{\beta}_k (x_{ki} - \bar{x}_k).$$

also, the baseline SHAP value ϕ_0 defaults exactly to the sample mean of the predictions, $\mu_{\hat{y}}$.

Substitute this linear SHAP identity back into our proposed tree-based decomposition formula for the absolute contribution:

$$D_k^{\text{SHAP}} = \frac{2}{n\mu_{\hat{y}}} \sum_{i=1}^{n} \phi_{ik} \left(R_i - \frac{1}{2} \right).$$

$$D_k^{\text{SHAP}} = \frac{2}{n\mu_{\hat{y}}} \sum_{i=1}^{n} \left[\hat{\beta}_k (x_{ki} - \bar{x}_k) \right] \left(R_i - \frac{1}{2} \right).$$

Because $\bar{x}_k$ is a constant and $\sum_{i=1}^{n}(R_i - \frac{1}{2}) = 0$, the mean-centering term drops out of the summation:

$$D_k^{\text{SHAP}} = \hat{\beta}_k \left[\frac{2}{n\mu_{\hat{y}}} \sum_{i=1}^{n} x_{ki} \left(R_i - \frac{1}{2} \right) \right].$$

By definition, the concentration index of determinant k is $C_k = \frac{2}{n\bar{x}_k} \sum_{i=1}^{n} x_{ki}(R_i - \frac{1}{2})$. Multiplying and dividing by $\bar{x}_k$, we rewrite the bracketed term:

$$D_k^{\text{SHAP}} = \hat{\beta}_k \left(\frac{\bar{x}_k}{\mu_{\hat{y}}} \right) C_k.$$

This is identical to the Wagstaff contribution formula(Wagstaff, van Doorslaer and Watanabe, 2003; Speybroeck *et al.*, 2010). This means the SHAP-based decomposition generalises the standard approach, retaining equivalence under linearity while expanding to accommodate tree ensembles.

3.5 Simulation

We ran a small simulation study to check whether the algorithm behaved as expected. We first generated covariates: residence, maternal education, and province. Then, we produced two datasets. The null data set and the positive control. This was determined by wealth and under-5 mortality distribution. In the null case, wealth and mortality were generated randomly. In the positive case, the planted structure comprised three variables: residence, maternal education, and province. Rural children formed the main at-risk group. Within this group, children in Kasai and Nord-Kivu formed a higher-risk subgroup compared with those in Kinshasa and Kongo Central. Children with no maternal education formed the highest-risk subgroup. These planted groups were assigned lower wealth and higher under-five mortality risk, with the strongest increase among poorer children in the highest-risk subgroup.

3.6 Analysis

We divided the analysis into direct and indirect decomposition methods. The direct methods use concentration-index trees and forests, developed here, in which the model is built to reduce socioeconomic inequality in under-five mortality at each split. In these methods, the subgroup structure is part of the decomposition because the tree is fitted using the concentration-index criterion. The indirect methods first fit a model for under-five mortality and then decompose the fitted mortality pattern across the socioeconomic ranking.

For the regression-based indirect decomposition, we fitted a survey-weighted logistic regression with a quasibinomial family using the survey R package (Lumley, 2026). The model was fitted using the DHS survey design object, including the sampling weights and the available clustering and stratification variables. This allowed the regression model to account for the DHS sampling design when estimating the association between under-five mortality and the observed child, maternal, household, occupational, and regional determinants. We then used the margins R package to obtain average marginal effects from the fitted survey-weighted model (Thomas J.

Leeper, 2024). These average marginal effects express the average change in predicted under-five mortality associated with each determinant on the response scale. We then passed these marginal effects, together with the model matrix, the DHS wealth ranking variable, the under-five mortality outcome, and the sampling weights, into the Rineq R package to compute the concentration-index decomposition (Wagstaff, van Doorslaer and Watanabe, 2003; Speybroeck *et al.*, 2010; Devleesschauwer *et al.*, 2025).

The second indirect method is the SHAP-based decomposition. For this method, we fitted a predictive random forest using the tidymodels R package with the ranger R package as the modelling backend (Wright and Ziegler, 2017; Kuhn and Wickham, 2020). Because under-five deaths were rare relative to surviving children, the alive class was undersampled during model fitting to reduce class imbalance. The resulting forest was therefore used as a predictive scoring model rather than as a calibrated probability model. We then used the shapr R package to estimate SHAP contributions from the fitted forest (Jullum *et al.*, 2025). The SHAP analysis was used to describe how the fitted forest used the predictors, not to estimate causal or population-level contributions of predictors to under-five mortality.

We also fitted a comparison concentration-index tree using the rpart R package with a custom concentration-index splitting method (Therneau and Atkinson, 2026). The comparison used the same outcome, socioeconomic ranking variable, and analysis weights as the main concentration-index tree. Node impurity was also defined using the absolute concentration index. The main difference was in the tree-growing tools. In the main implementation, we used custom tree-growing controls and the partykit R package for tree representation and plotting (Hothorn and Zeileis, 2015). A split was accepted only if it satisfied the concentration index tree stopping rules described in the tree the tree stopping rules section. In the rpart comparison, split search and stopping were controlled via rpart controls, primarily minsplit, minbucket, and the complexity parameter cp.

For the concentration-index trees and forests, validation folds were created at the DHS primary sampling unit level. All children from the same PSU were kept in the same fold. The model was fitted on the training PSUs and then evaluated on held-out PSUs using the same validation-gain measure. This was done because children in the same survey cluster may share local conditions and should not be treated as fully independent observations during validation. This follows the idea of blocked cross-validation for structured data (Roberts *et al.*, 2017). We report the training gain, validation gain, relative validation gain, and tree size for the selected setting. We performed a stability check of gain and relative gain using a bootstrap. All children from the sampled PSUs were retained, and the tree was refitted using the selected tuning settings using sampling with replacement. The bootstrap results were used to describe how often the same split variables and similar terminal-node structures were recovered, not as a formal replicate-weight survey inference procedure. Tree similarity was measured using Jaccard overlap between terminal-node memberships in the selected tree and the bootstrap-refitted tree. A higher Jaccard value means that the two trees placed more of the same children into comparable terminal subgroups.

4 Results

4.1 Exploratory Data Analysis

In the DRC Data Set(The DRC Program Program, 2007), 10.32% of children died before reaching age five (Figure 3). Deaths were more commonly observed among children from disadvantaged or rural backgrounds. The children who died differed descriptively from those who survived. Low-wealth households accounted for a larger share of deaths than of survivors (51.3% versus 42.8%). The same pattern was present for rural residence (69.3% versus 60.7%), no skilled attendance at birth (65.6% versus 56.8%), and mothers with no education (67.5% versus 55.2%) (Table 3). Mortality also differed across provinces, ranging from 6.2% in Kinshasa to 14.2% in Katanga (Figure 15). Next we use the direct and indirect methods for decomposition.

Figure 3: Under-five mortality distribution and DHS wealth index in the DRC analysis data.

4.2 Decomposition of determinants of under-five mortality

4.2.1 Indirect methods comparison (regression and predictive-forest SHAP)

From the indirect decomposition, the largest CI regression contribution was mother's education, followed by father's occupation in agriculture, mother's occupation in agriculture, household wealth, residence, and skilled birth attendance (Table 1). In the predictive-forest SHAP decomposition, the contributions were less concentrated in the education and occupation variables. Residence gave the largest SHAP contribution under CI, CIg, CIc, and L. Under CI, CIg, and CIc, residence contributed about 19%, followed by father's agricultural occupation, household wealth, mother's education, Maniema, mother's agricultural occupation, and skilled birth attendance. Under L, residence was again the largest contributor, followed by father's agricultural occupation, mother's education, Maniema, mother's agricultural occupation, Kinshasa, and household wealth. The fitted regression model is reported in Table 4. Details of the fitted predictive forest are provided in Table 5 and Table 6, and its SHAP contributions are shown in Figure 16.

Table 1: Regression and predictive-forest SHAP percentage contributions by criterion.

Variable	Regression			Predictive-forest SHAP			
	CI (%)	CIg (%)	CIc (%)	CI (%)	CIg (%)	CIc (%)	L (%)
Mother's education	31.232	170.974	170.974	8.101	8.101	8.101	9.518
Father's occupation: Agriculture	24.119	117.979	117.979	8.537	8.537	8.537	9.633
Mother's occupation: Agriculture	16.135	87.330	87.330	6.864	6.864	6.864	6.730
Household wealth	12.699	53.789	53.789	8.335	8.335	8.335	6.155
Type of residence	12.019	71.760	71.760	19.418	19.418	19.418	23.992
Skilled birth attendance	10.741	60.069	60.069	6.415	6.415	6.415	1.362
Region: Equateur	5.118	6.371	6.371	4.116	4.116	4.116	3.327
Father's occupation: Household, unskilled, not working	-4.569	-5.692	-5.692	-0.667	-0.667	-0.667	-1.049
Region: Kinshasa	-4.102	-3.621	-3.621	3.204	3.204	3.204	6.102
Region: Sud-Kivu	-2.783	-1.299	-1.299	-3.520	-3.520	-3.520	-4.137
Region: Kasai Oriental	-2.287	-2.951	-2.951	1.548	1.548	1.548	1.192
Father's education	-2.023	-5.285	-5.285	2.185	2.185	2.185	3.424
Region: Maniema	1.604	0.547	0.547	7.644	7.644	7.644	8.955
Region: Orientale	-1.174	-1.314	-1.314	-0.545	-0.545	-0.545	-0.377
Birth order and interval: 5+ short interval	-0.998	-0.985	-0.985	-0.105	-0.105	-0.105	4.641
Region: Bas-Congo	-0.926	-0.360	-0.360	0.535	0.535	0.535	0.749
Mother's occupation: Household, unskilled, not working	0.778	1.749	1.749	-0.748	-0.748	-0.748	-1.236
Region: Kasai Occidental	0.629	0.648	0.648	-0.821	-0.821	-0.821	-0.259
Birth order and interval: 5+ long interval	-0.615	-1.565	-1.565	-1.987	-1.987	-1.987	-2.629
Sex of the child	-0.581	-2.775	-2.775	-1.638	-1.638	-1.638	0.343
Region: Katanga	-0.447	-0.446	-0.446	2.125	2.125	2.125	2.451
Mother's age at birth	0.364	1.522	1.522	0.651	0.651	0.651	-1.015
Region: Nord-Kivu	0.233	0.088	0.088	1.128	1.128	1.128	1.094
Birth order and interval: 2-4 long interval	0.208	0.685	0.685	-1.104	-1.104	-1.104	-2.907
Birth order and interval: 2-4 short interval	-0.006	-0.006	-0.006	-6.310	-6.310	-6.310	-6.461

4.3 Direct methods (concentration-index trees and forests)

4.3.1 Rank Dependent Concentration-Index Tree

The Corrected Concentration-Index Tree CIc

We start with a tree based on the corrected concentration index, appropriate for bounded outcomes (Erreygers, 2009). In the fitted data, the selected CIc tree shows that inequality in under-five mortality is structured mainly through maternal education, father's agricultural occupation, skilled birth attendance, household wealth, and residence in Katanga Figure 4. The first separation is by maternal education. Among children whose mothers had some education, the tree then separates the father's agricultural occupation and skilled birth attendance. Among children whose mothers had no education, the tree separates household wealth and then Katanga. The lowest-mortality terminal group consists of children whose mothers had some education and whose fathers were not in agriculture, with an under-five mortality rate of about 6.5%. The highest-mortality terminal group consists of children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga, with a mortality of about 17.8%.

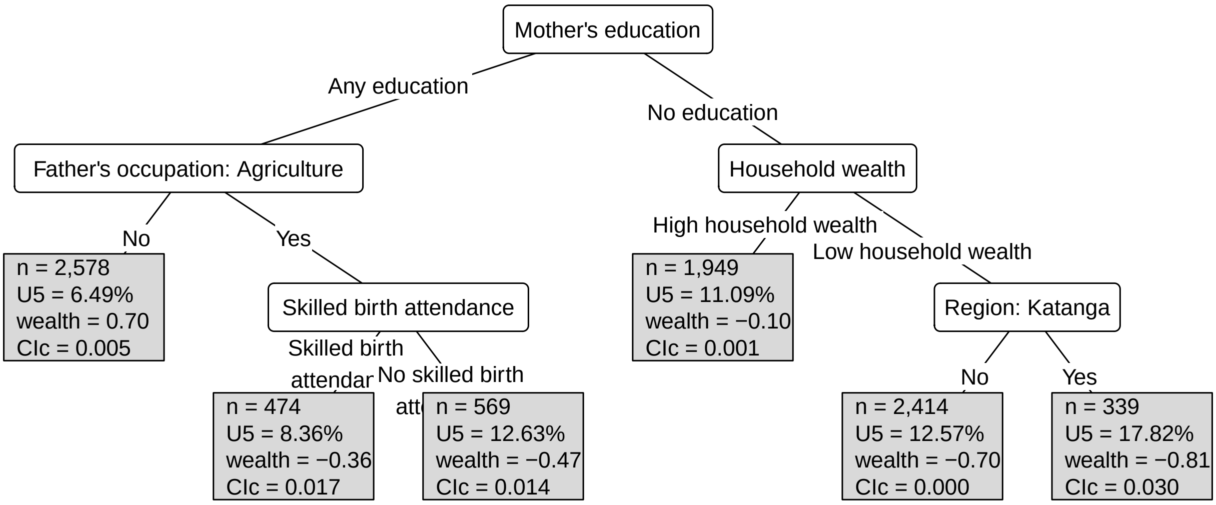

Figure 4: Concentration-index tree fitted with the CIc criterion.

Generalised and standard concentration-index trees CIg and CI

The appendix also shows the trees fitted using the generalised and standard concentration indices. The generalised concentration-index tree gives the same fitted subgroup structure as the corrected concentration-index tree Figure 18. The mortality pattern is therefore substantively unchanged.

The standard concentration-index tree gives a closely related structure, but it adds one further split among children whose mothers had some education and whose fathers were not in agriculture Figure 19. In this branch, children without a short preceding birth interval have the lowest mortality, about 6.0%, while children with a short preceding birth interval have a mortality of about 11.7%. Apart from this additional demographic distinction, the standard CI tree points to the same main sources of inequality as the corrected and generalised trees: maternal education, father's agricultural occupation, skilled birth attendance, household wealth, and residence in Katanga.

4.3.2 Level dependent Concentration-Index Tree L

We then fit a tree based on the level-dependent concentration index criterion, L (Erreygers and Kessels, 2017). This index is most appropriate when the socioeconomic variable has ratio-scale properties. We treat it as a sensitivity analysis; the selected L tree shows that inequality in under-five mortality is structured primarily by rural residence, father's agricultural occupation, household wealth, and province, especially Kinshasa, Equateur, and Katanga Figure 5. The lowest-mortality terminal group consists of urban children in Kinshasa whose fathers were not in agriculture, with an under-five mortality rate of about 5.1%. The highest-mortality terminal group consists of rural children living in low-wealth households in Katanga, with an under-five mortality rate of about 18.5%. Thus, the level-dependent tree tells a more spatial story of inequality: mortality risk is first organised by the rural-urban divide, then by occupation, wealth, and region. This structure differs from the rank-dependent trees, which first organise the fitted subgroups around maternal education and father's agricultural occupation.

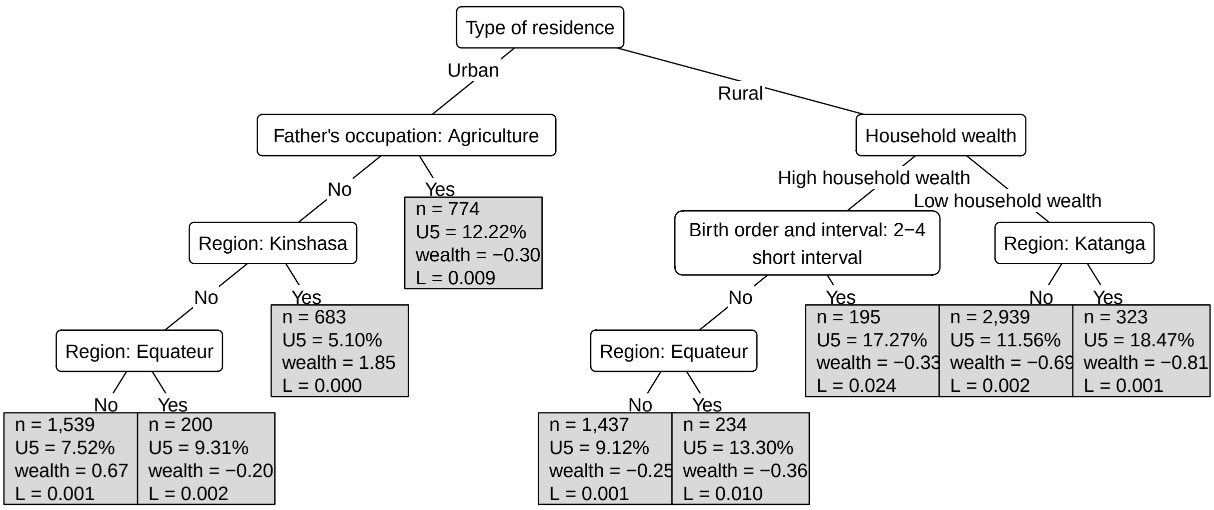

Figure 5: Level dependent concentration-index tree for under-five mortality in the DRC.

4.3.3 Tree validation plots

The validation results indicated that the subgroup structure learned by these trees during training did not generalise well to held-out samples. The rank-dependent trees based on CI, CIg, and CIc gave strong training gains, removing 83.3% of root-node impurity for CI and 81.6% for both CIc and CIg Table 8. However, their validation gains were negative under the PSU-held-out folds Figure 6, Table 8. A negative validation gain indicates that the fitted terminal nodes did not reduce the concentration-index impurity in the held-out PSUs relative to the root node. Thus, although the rank-dependent trees identified plausible subgroup structure in the fitted sample, the same partitions did not generalise well to clusters not used to fit the tree.

The level-dependent L tree showed a different validation pattern. It removed 98.2% of root-node impurity during training. It retained a positive validation gain of 64.5% under the same PSU-held-out validation criterion, so the rural-urban, occupational, wealth, and regional partition was more reproducible across held-out clusters Figure 6, Table 8. This does not make L the definitive main result, because L is most appropriate when the socioeconomic variable has ratio-scale properties, and the DHS wealth index is a PCA-derived asset score.

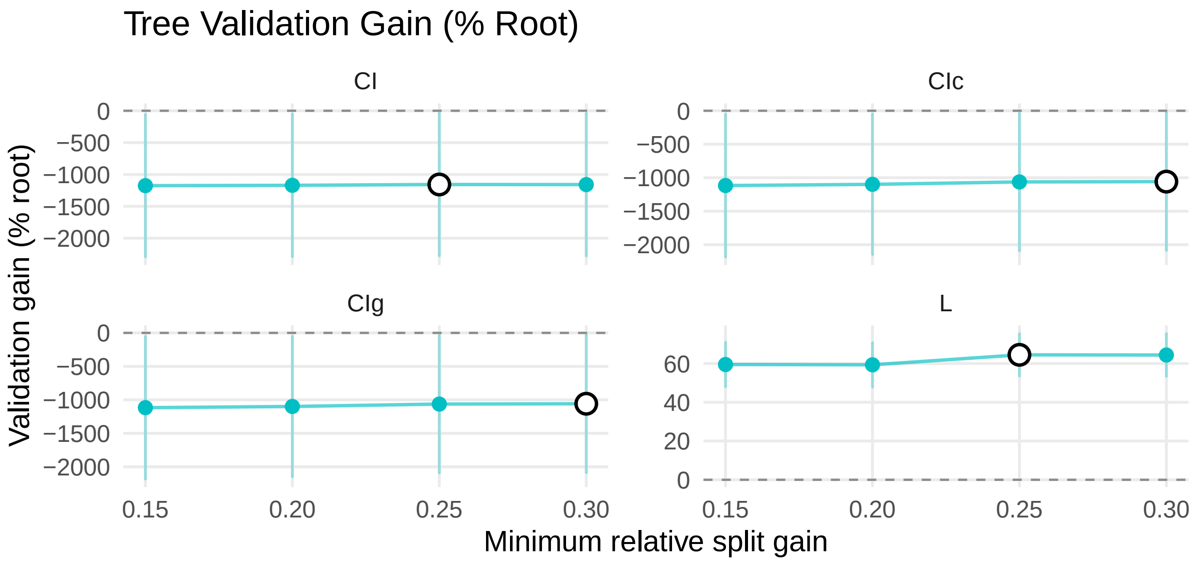

Figure 6: Tree validation gain and complexity by minimum relative split gain.

Tree complexity plots

Tree complexity decreased as the minimum relative split gain increased, since larger thresholds allowed only splits that removed a larger share of parent-node impurity Figure 7. The selected settings produced relatively small trees across the 10 PSU-held-out validation folds, with mean terminal-node counts of 6.9 for CI, 6.0 for CIc, 6.0 for CIg, and 7.5 for L Table 9, Figure 7. These decimal values are averages across validation-fold refits. They should not be read as the number of terminal nodes in a single final fitted tree, which must be an integer.

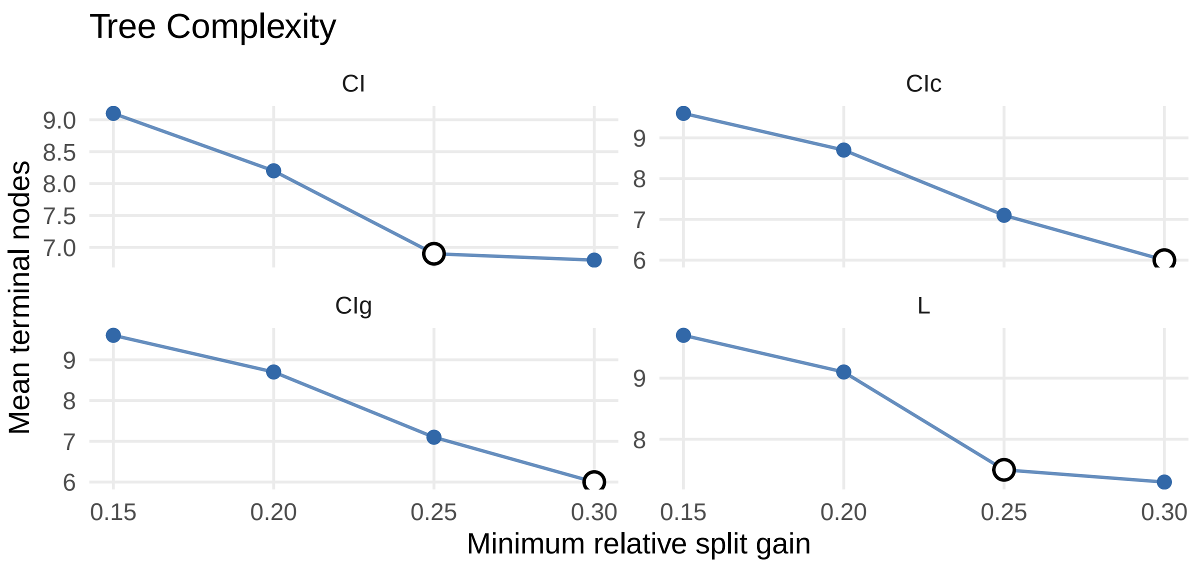

Figure 7: Tree complexity by number of terminal nodes.

Bootstrap stability of selected trees

Table 2: Bootstrap stability of selected concentration-index trees

Criterion	No split refits	Terminal nodes	Root split	Relative gain	Tree similarity
CI	1.0%	6 (2, 10)	Mother's education, 41.7%	0.463 (0.287, 0.767)	0.270 (0.074, 0.683)
CIg	3.0%	5 (1, 9)	Mother's education, 39.9%	0.500 (0.330, 0.789)	0.313 (0.148, 0.758)
CIc	1.8%	5 (2, 9)	Mother's education, 45.3%	0.505 (0.335, 0.800)	0.331 (0.148, 0.761)
L	0.0%	6 (3, 11)	Type of residence, 67.2%	0.930 (0.784, 0.997)	0.277 (0.074, 0.841)

The bootstrap stability checks resampled primary sampling units and refitted the selected tree settings. All criteria completed 1000 successful refits with no failures. The table shows that the rank-dependent trees most often split first on mother's education, while the L tree most often split first on residence. The L tree had the highest relative gain, but tree similarity remained modest across all criteria.

Rpart comparison

The rpart comparison tree recovered the same upper-level structure as the main CI tree (Figure 19; Figure 8). Both trees first split on maternal education, then separated children by father's agricultural occupation among those whose mothers had some education, and by household wealth among those whose mothers had no education. The high-mortality branch was also similar, identifying children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga. The main difference was in the lower branches: the rpart comparison introduced a split by long preceding birth interval, while the main CI tree used a short preceding birth-interval split. This suggests that the main rank-dependent structure was not driven only by the custom implementation, although some lower-level refinements depended on the tree-growing and stopping rules.

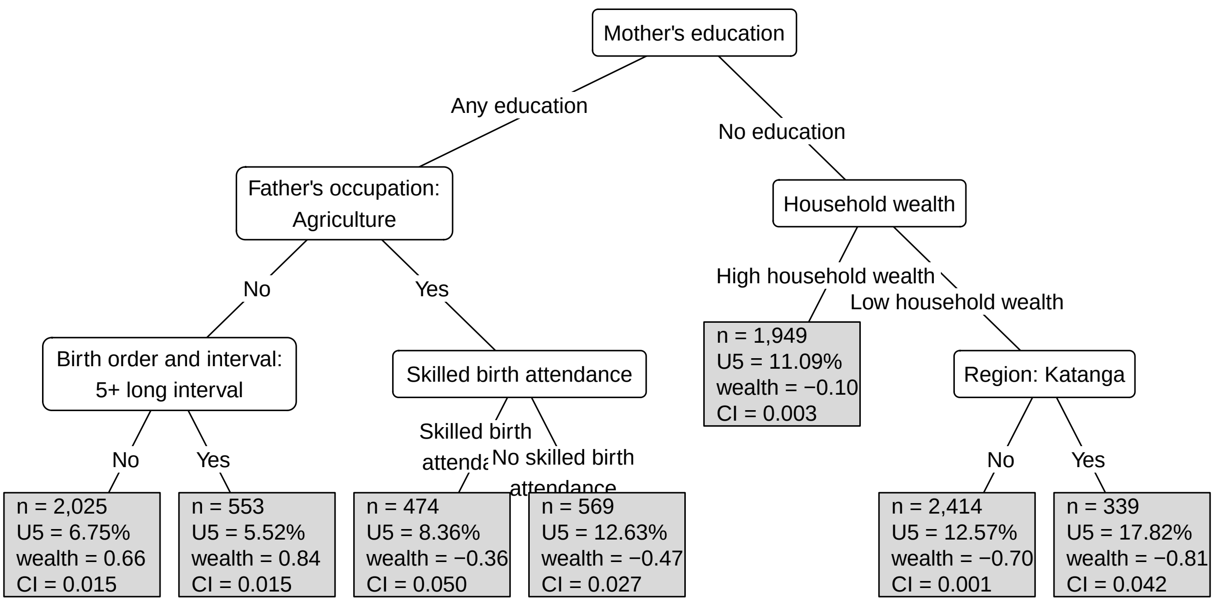

Figure 8: Rpart comparison tree using CI impurity.

4.3.4 Concentration index forest and surrogate tree

Forest validation plots

We evaluated concentration-index forests using the same impurity-reduction validation criterion as the single trees. The selected forest settings used a minimum relative split gain of 0.20 to 0.25; larger values produced very shallow forests with mean terminal-node counts close to the root, so they did not reveal additional subgroup structure Figure 9, Figure 21.

For rank-dependent criteria, forests reduced training gains relative to corresponding single trees but did not yield a stable validation pattern. The CI, CIc, and CIg forests removed 48.1%, 51.7%, and 51.6% of root-node impurity in training Table 11, Table 8. Their validation-gain ratios remained negative: -1038.3% for CI, -1108.5% for CIc, and -1063.0% for CIg Figure 9, Table 11. Thus, averaging across trees reduced the size of the training gain, but it did not make the rank-dependent forest partitions reproducible under the validation criterion.

The tuning path shows the trade-off between validation performance and complexity. As the minimum relative split gain increased, the component trees became shallower Figure 9, Figure 21. The level-dependent forest remained the most stable. It removed 95.3% of root-node impurity in training and retained a positive validation gain of 56.7% Table 11. As with the L tree, this result is interpreted as a sensitivity analysis because the DHS wealth index is not a ratio-scale socioeconomic variable.

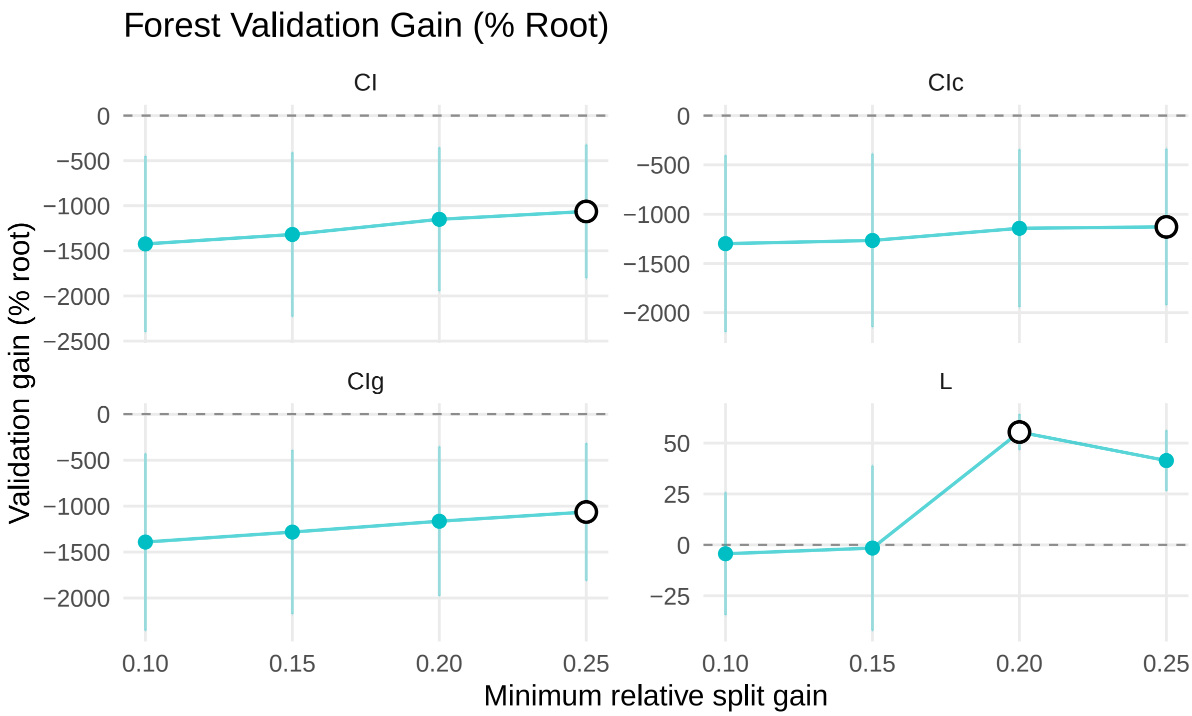

Figure 9: Forest validation gain and complexity by minimum relative split gain.

Surrogate trees from Forest Predictions

To interpret the selected forest, we fitted a surrogate tree to summarise its predictions (Figure 10). The surrogate tree retained the main structure seen in the CIg forest. We present the CIg surrogate because the CIc surrogate did not identify terminal nodes with concentration-index values close to zero.

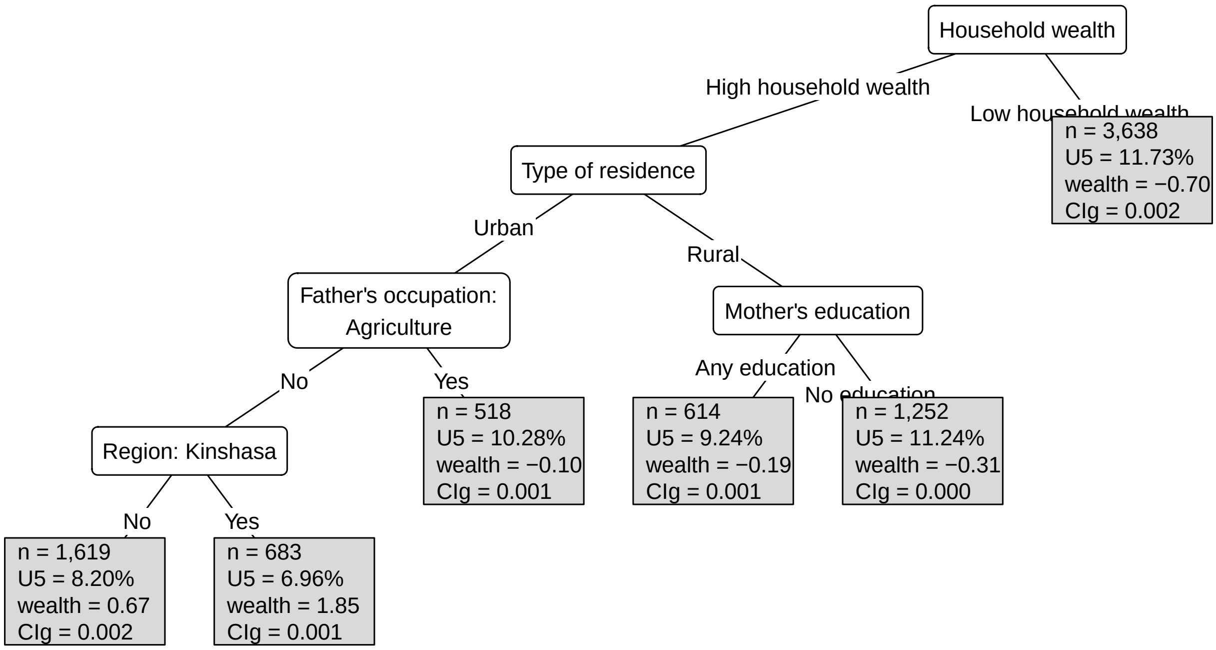

Figure 10: Surrogate tree for the CIg concentration-index forest.

The other surrogate trees can be found in the appendix (Figure 26; Figure 25; Figure 23).

4.3.5 Variable importance

The variable-importance plots compare the fitted CIc tree with the CIg forest that is summarised by the displayed surrogate tree Figure 11. In the CIc tree, the largest reductions in concentration-index impurity come from mother's education, father's agricultural occupation, household wealth, Katanga, and skilled birth attendance.

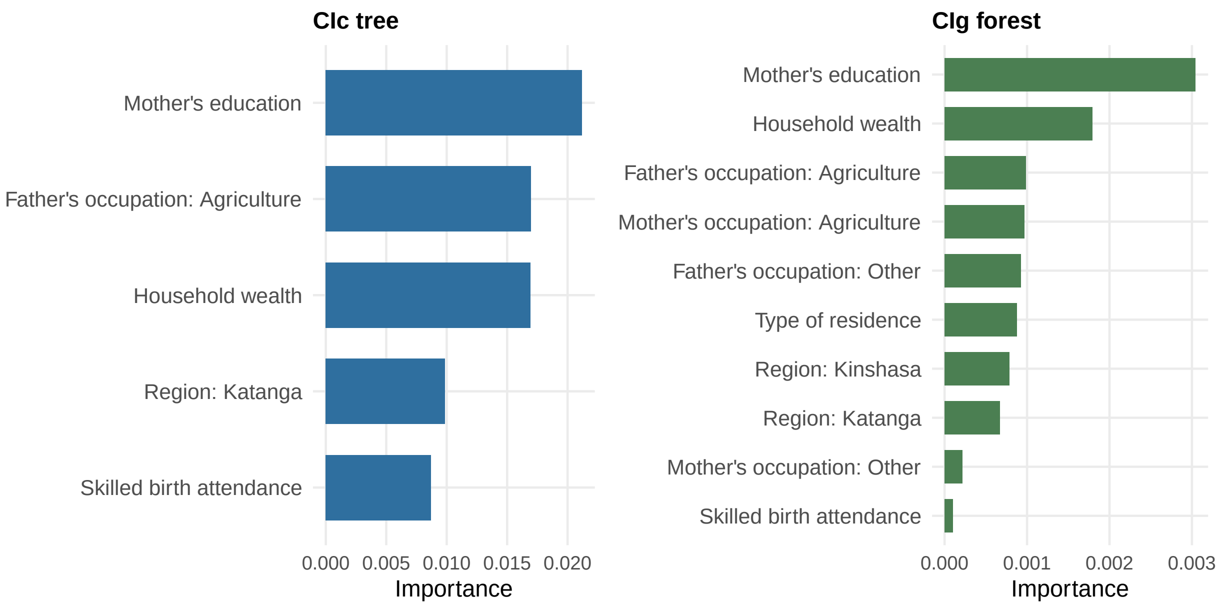

Figure 11: Variable importance for the CIc concentration-index tree and CIg concentration-index forest.

The CIg forest gives a broader ranking than the single CIc tree. Mother's education remains the leading variable, but the forest also gives weight to household wealth, father's agricultural occupation, mother's agricultural occupation, residence, and regional indicators. In the appendix plots, the same groups of variables appear again under the other criteria: residence, region, occupation, education, and wealth Figure 17, Figure 22.

4.4 Simulation Study

4.4.1 Null case

From (Figure 12) we see that the algorithm developed here did not find a meaningful structure and behaved as expected. This shows that the tree-growing rule does not create subgroup structure when the outcome and socioeconomic ranking are independent, since the tree uses concentraion index as an impurity measure, which measures how sociaeconomic ranking and outcome covary.

(a) CI (b) CIg

(c) CIc (d) L

Figure 12: Null simulation concentration-index trees by criterion.

4.4.2 Poor subgroup case

In the positive-control simulation, the method(the concentration index tree with CIc as an impurity measure) recovered the planted subgroup structure. The tree first separated urban from rural observations, then split the rural group by province and maternal education (Figure 13). This structure matched the way the simulated outcome was generated: poorer observations in the planted rural, low-education, selected-province subgroup had higher outcome prevalence. The terminal nodes also followed the expected gradient, with the highest prevalence among observations in the most disadvantaged planted subgroup.

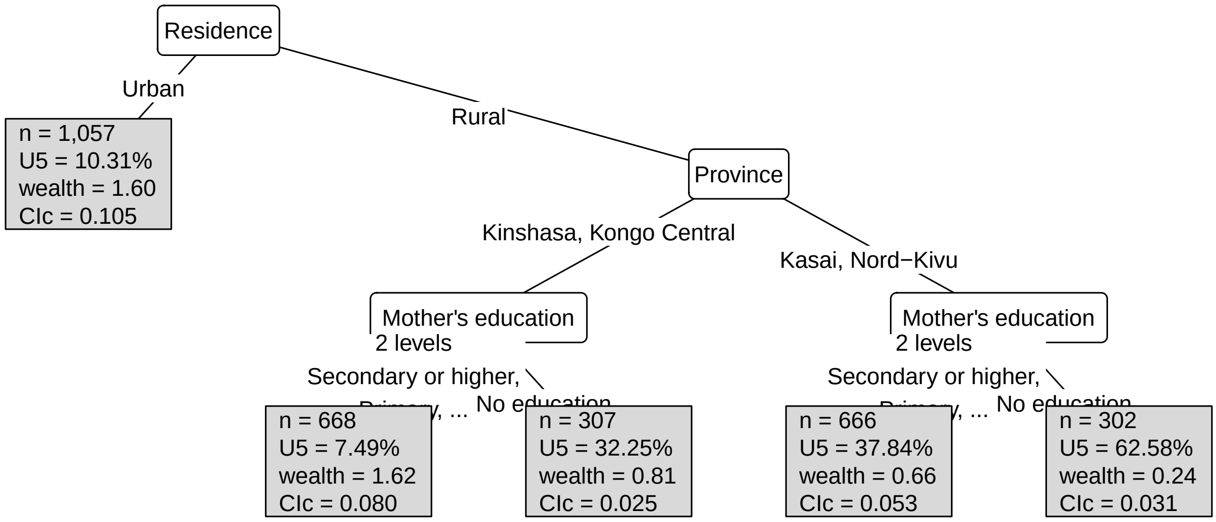

Figure 13: CIc concentration-index tree recovered from simulated data with a known subgroup-generating mechanism.

We see the same pattern across the sensitivity plots in the appendix the trees developed using CI, CIg, and CIc, and L as impurity measures(Figure 27; Figure 28; Figure 29).

5 Discussion

5.1 Main findings

This thesis developed and applied a tree-based model for analysing socioeconomic inequality in under-five mortality using the concentration index impurity. The main finding is that inequality in the DRC data could not be explained by household wealth alone. Under-five mortality was also related to residence, maternal education, skilled birth attendance, parental occupation, and province. Overall, 10.32% of children died before age five. The children who died were more often from low-wealth households, rural households, mothers with no education, births without skilled attendance, and some provinces, especially Katanga. This is in line with earlier DRC work on geographic variation in under-five mortality, and with decomposition studies showing that wealth, education, residence, and occupation contribute to mortality inequalities in African settings (Van Malderen, Van Oyen and Speybroeck, 2013; Kandala *et al.*, 2014; Van Malderen *et al.*, 2019).

The indirect methods provided a variable view of this inequality. In the regression based decomposition, mother's education made the largest regression contribution, followed by father's agricultural occupation, mother's agricultural occupation, household wealth, residence, and skilled birth attendance. These results are consistent with earlier decomposition work, which found that household wealth, parental education, occupation, and residence were important contributors to under-five mortality inequality (Van Malderen, Van Oyen and Speybroeck, 2013; Van Malderen *et al.*, 2019). The SHAP analysis, from the fitted random forest, gave a less concentrated picture. Residence was the largest SHAP contributor, while household wealth, parental occupation, mother's education, skilled birth attendance, and some regional indicators also appeared across criteria. SHAP-based decomposition therefore adds to the regression methods by allowing a more flexible fitted model, such as a random forest, to be summarised at the determinant level.

These SHAP results should be read as an explanation of the fitted predictive model rather than as causal or population-level determinant contributions.

The direct concentration-index trees added the subgroup interpretation that is less visible in the regression-based decomposition. While the indirect methods ranked variables by their contribution to inequality, the trees showed how these variables combined within the data to define groups with different inequality patterns. The rank-dependent trees based on CI, CIg, and CIc identified a structure involving maternal education, father's agricultural occupation, skilled birth attendance, household wealth, and Katanga. Mortality was lower among children whose mothers had some education and whose fathers were not in agriculture, and higher among children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga. However, these partitions did not validate well. Although the rank-based methods removed about 87% to 88% of root-node inequality in training, their validation gains were negative, suggesting that the strong training structure did not transfer well to held-out folds.

5.2 Rank-dependent and level-dependent views of inequality

From these analyses, we see differences in the results (Figure 19; Figure 18; Figure 4) in subgroups formed by the two ways of defining the impurity measure when developing the concentration index tree. In the rank-dependent criteria CI, CIg, and CIc, a child's contribution depends on the child's position in the current node's socioeconomic ranking. After each split, the children in the new subgroup are ranked again, so the same child can receive a different rank because the comparison group has changed. In the level-dependent criterion, re-ranking is still done but in L, the child's observed socioeconomic level is carried into the subgroup, so a child's rank also reflects their wealth level.

The differences also showed up in cross-validation; the level-dependent method outperformed the rank-dependent methods. The rank-dependent trees had strong training performance but negative validation gains, while the level-dependent tree kept a positive validation gain (Figure 6; Table 8). The same pattern was seen in the forests: averaging trees reduced the validation loss for the rank-dependent criteria, but their validation gains remained below zero, whereas the L forest retained a positive validation gain (Figure 9; Table 11). The validation results show that CI, CIg, and CIc identified detailed in-sample subgroup rules, whereas L produced a more reproducible subgroup structure in this application.

These results are also mirrored by the subgroup-decomposability argument in (Erreygers *et al.*, 2017). In their comparison of rank-dependent and level-dependent measures, they show that the level-dependent index L is subgroup-decomposable and rank-dependent measures are more difficult to decompose cleanly across population groups. Decomposability of the level-dependent concentration index shown by (Erreygers *et al.*, 2017) can also explain why L performed better as a tree impurity criterion in this application: children inherit their socioeconomic level within each subgroup, rather than being re-weighted only by a new within-node rank. Although the DHS wealth index is an asset-based score(The DHS Program, 2007) and (Erreygers and Kessels, 2017) recommends using Level depedent concentration index(L) with a wealth-ranking variable that has ratio properties. The cross-validation results show that the L concentration index impurity measure yielded a structure transferable to held-out samples using the DHS wealth score.

5.3 Methodological and data limitations

This thesis uses DHS survey data from the Democratic Republic of the Congo (The DRC Program Program, 2007). The DHS Program states that participation is voluntary and that respondents' identities and information are kept confidential (The DHS Program, 2026). However, the data still have the usual limitations of retrospective household surveys. Under-five mortality is reconstructed from birth histories, so the estimates depend on correct reporting of births, dates, and child survival status. Non-response, recall error, and misreporting of dates may affect the mortality estimates. The survey is also from 2007. It was useful for comparing the new method with earlier work on the same setting. However, the results should not be read as describing the current situation in the Democratic Republic of the Congo.

The second limitation is the socioeconomic ranking variable. The DHS wealth index is constructed from household assets and living-standard variables, typically using principal component analysis (The DHS Program, 2007). It is useful for ordering households from poorer to richer, but it is not a measure of income, consumption, or expenditure. It is also a relative score within the survey. This limitation mainly affects the level-dependent concentration index. For L, the interpretation is clearer when the socioeconomic variable has ratio-scale properties (Erreygers and Kessels, 2017).

Decision trees are known to be unstable: small changes in the data can produce different split variables, split points, and tree structures. This comes from the recursive and greedy nature of tree construction. Once an early split changes, all later splits are fitted to different subsets of the data. In this thesis, this was handled partly through cross-validation and through splitting controls such as minimum node size, minimum child-node proportion, maximum depth, minimum gain, and minimum relative gain. These controls helped restrict weak or very small splits, but they do not remove the instability problem completely.

Another limitation is the use of a surrogate tree to summarise the forest. The forest was added to reduce dependence on a single greedy tree, but the resulting forest is harder to interpret than a single tree. To obtain an interpretable summary, a surrogate tree was fitted using the forest's risk estimates. The surrogate still chooses splits by maximising concentration-index gain, but the outcome used in that split rule is the forest-predicted risk rather than the observed mortality outcome. This approximation of the forest, therefore, depends on the correctness of the forest.

Another limitation comes from SHAP-based decomposition. This analysis was based on a predictive random forest and should be interpreted as an explanation of that fitted prediction model, not as a decomposition of population mortality risk. SHAP values depend on the fitted model and on the SHAP convention used. This is important in DHS data because maternal education, occupation, residence, and household wealth are correlated. The allocation of predictive contribution across these variables is therefore not unique, and the SHAP results should not be interpreted as causal or determinant contributions. The predictive forest used for the SHAP analysis was also affected by undersampling. As shown in the appendix, the alive class was undersampled, changing the weighted class distribution from about 89.7% alive and 10.3% died to about 60.0% alive and 40.0% died. Unless the predictions are recalibrated, the fitted values from this model should be treated as prediction scores rather than calibrated mortality probabilities. The predictive performance of this forest was modest, with ROC AUC values around 0.61. This does not invalidate the SHAP analysis, but it limits confidence in detailed feature rankings. For this reason, the SHAP results are treated as a descriptive explanation of the fitted predictive model, rather than as evidence that the listed variables contribute those proportions to the risk of under-five mortality.

5.4 Ethical, societal, and stakeholder relevance

This thesis uses child-level data from the Democratic Republic of the Congo DHS phase V survey (The DRC Program Program, 2007). These data contain sensitive information, including child survival status, birth histories, household wealth, residence, province, maternal education, occupation, and skilled birth attendance. Even without names, the data describe mothers, children, households, and family deaths. The DHS Program states that participation is voluntary and that respondents' identities and information are kept confidential (The DHS Program, 2026). In this thesis, we used secondary survey data and should remain within the conditions under which we were granted access to the DHS files.

The concentration index trees developed here can identify subgroups with higher mortality and greater socioeconomic disadvantage. For example, the fitted trees point to rural children from low-wealth households in Katanga, or children whose mothers had no education and who lived in low-wealth households. These subgroups are useful for public health analysis and planning, but the results should be used carefully. Groups we identified through concentration index trees should not be described as if the children, mothers, or provinces are the problem. The tree is showing where mortality and disadvantage are concentrated in the data. The interpretation should therefore focus on unequal conditions, access to services, and policy response, not labelling groups.

Compared with GLM based decomposition, tree-based models are easier to interpret. The variable percentage contribution from regression or Shap-based decomposition can show which variables contribute more. However, it cannot show how disadvantage is organised across residence, household wealth, occupation, maternal background, and province. A tree is easier to discuss because it works like a flow chart. As shown in this thesis, the tree partitions the data graphically into high- and low-risk groups. Concentration index trees make it easier to follow the path to these high- and low-risk groups and identify them for public health planning. This makes tree models useful in public health and policy contexts, where the goal is to identify groups for targeted interventions. They are also easy to interpret for non-technical groups.

6 Future research

In this thesis, the level-dependent L method yielded more stable validation results than the rank-dependent criteria. This suggests that the choice of concentration index, used here as the tree impurity measure, is an important part of the method. One useful direction for future work would be to investigate whether rank-dependent concentration-index trees can be modified so that a child node retains more information about the wealth position from which its observations came, rather than relying solely on the new within-node ranking after each split. This may help reduce some of the instability seen in the rank-dependent trees.

7 Conclusion

This thesis developed and applied a tree-based method for analysing socioeconomic inequality in under-five mortality. The method used the concentration index as the impurity measure in a recursive partitioning method. Instead of ranking determinants by their percentage contribution to inequality, the fitted tree produced subgroups that could be interpreted as decision rules. This added a subgroup view to the usual regression-based decomposition. In the DRC DHS application, inequality in under-five mortality was not explained solely by household wealth. Residence, maternal education, skilled birth attendance, parental occupation, household wealth, and province all appeared in the analysis.

The indirect methods gave the variable-level view, while the direct trees showed how these variables combined. The regression decomposition emphasised the mother's education and the father's agricultural occupation. The SHAP-based decomposition gave a broader pattern, with residence, household wealth, mother's education, skilled birth attendance, and parental occupation appearing across criteria.

The concentration-index trees acted as a direct method to decomposition, which showed directly with subgroups had higher under 5 deaths concentration. The rank-dependent trees found an education-occupation-wealth pattern, with higher mortality among children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga. The L method gave a more spatial structure, beginning with residence and then splitting by occupation, wealth, and province.

In this thesis, the level-dependent L method yielded more stable validation results than the rank-dependent criteria. This suggests that the choice of concentration index, used here as the tree impurity measure, is an important part of the method. The rank-dependent trees found detailed structures in the fitted data, but these structures did not validate well. The L tree and forest retained positive validation gains, so the level-dependent result was the more stable descriptive partition in this application.

Since the level-dependent concentration index, as a measure of impurity, produced better results on held-out folds, future research should examine how to improve rank-dependent trees to retain more information from the original wealth ranking after each split.

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9 Generative AI Disclosure

- I have used ChatGPT (GenAI tool) to support the development of the ineqTrees package after defining the main functions and writing starter code.
- I have used ChatGPT and Grammarly (GenAI/writing support tools) for English language improvement, spelling, and clarity.
- I used ChatGPT to help develop custom tree plots, especially plots where each node displays wealth, sample size, and the outcome.

10 Appendix

10.1 Code

- The Analysis code for this thesis is available at https://github.com/m-mburu/thesis-mse
- The code for the concentration-index tree and forest implementation is available at https://github.com/m-mburu/ineqTrees

10.2 Appendix A: Proof that the Centered Ranks Sum to Zero

This appendix provides the algebra used in Section 4.4.

For the standard fractional rank,

$$R_i = \frac{i - 0.5}{n},$$

where i is the individual's position (from 1 to n) when ordered by socioeconomic status.

If we sum this rank across all n individuals, we obtain

$$\begin{aligned}\sum_{i=1}^{n} R_i &= \frac{1}{n}\sum_{i=1}^{n}(i-0.5)\\ &= \frac{1}{n}\left(\sum_{i=1}^{n} i - \sum_{i=1}^{n} 0.5\right).\end{aligned}$$

Using $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ and $\sum_{i=1}^{n} 0.5 = 0.5n$, it follows that

$$\begin{aligned}\sum_{i=1}^{n} R_i &= \frac{1}{n}\left(\frac{n(n+1)}{2} - 0.5n\right)\\ &= \frac{1}{n}\left(\frac{n^2+n-n}{2}\right)\\ &= \frac{1}{n}\left(\frac{n^2}{2}\right)\\ &= \frac{n}{2}.\end{aligned}$$

Therefore,

$$\begin{aligned}\sum_{i=1}^{n}\left(R_i - \frac{1}{2}\right) &= \sum_{i=1}^{n} R_i - \sum_{i=1}^{n}\frac{1}{2}\\ &= \frac{n}{2} - \left(n \times \frac{1}{2}\right)\\ &= \frac{n}{2} - \frac{n}{2}\\ &= 0.\end{aligned}$$

10.3 Appendix B: Numerical Example Comparing CI and CIg

This appendix illustrates the result stated in Section 3.3.4.

Consider two candidate splits s_1 and s_2 with equal child-node weights $p_L = p_R = \frac{1}{2}$. Here A_{t_L} and A_{t_R} denote the absolute generalised concentration-index components in the left and right child nodes, while μ_{t_L} and μ_{t_R} are the corresponding child-node mean outcomes.

For s_1, suppose

$$A_{t_L} = A_{t_R} = 0.06, \qquad \mu_{t_L} = \mu_{t_R} = 0.10.$$

For s_2, suppose

$$A_{t_L} = A_{t_R} = 0.08, \qquad \mu_{t_L} = \mu_{t_R} = 0.50.$$

Then the CIg-based child-node objective is

$$\frac{1}{2}(0.06) + \frac{1}{2}(0.06) = 0.06 \quad \text{for } s_1,$$

and

$$\frac{1}{2}(0.08) + \frac{1}{2}(0.08) = 0.08 \quad \text{for } s_2.$$

So CIg prefers s_1.

However, the CI-based child-node objective is

$$\frac{1}{2}\left(\frac{0.06}{0.10}\right) + \frac{1}{2}\left(\frac{0.06}{0.10}\right) = 0.60 \quad \text{for } s_1,$$

and

$$\frac{1}{2}\left(\frac{0.08}{0.50}\right) + \frac{1}{2}\left(\frac{0.08}{0.50}\right) = 0.16 \quad \text{for } s_2.$$

So CI prefers s_2. This shows that relative and absolute inequality criteria can lead to different split choices.

10.4 Greedy Concentration-Index Tree-Growing Procedure Flowchart

Algorithmic Summary

The complete procedure we have built up to this point is summarised as follows.

Algorithm: Concentration-index-based recursive partitioning

Input:

- health outcome Y
- socioeconomic ranking variable S
- candidate predictors $X_1, \dots, X_p$
- survey weights w_i

At node t:

1. Compute the weighted within-node socioeconomic ranks $R_{i|t}$ from S.
2. Stop if maxdepth, the maximum permitted tree depth, or minsplit, the minimum weighted parent-node size, prevents further splitting.
3. Optionally sample mtry candidate predictors, where mtry is the number of predictors searched at a node.
4. For every candidate predictor X_j, generate binary split candidates:
    - if X_j is numeric, evaluate cutpoints s between distinct ordered values;
    - if X_j is categorical, evaluate admissible binary partitions of its levels, using ordered cumulative partitions where needed.
5. Reject candidates that violate minbucket, the minimum weighted child-node size, or minprob, the minimum child-node share of the parent-node weight.
6. For every remaining candidate pair (j, s), compute the concentration-index gain $G_m(j, s, t)$.
7. Choose the split

$$(j^*, s^*) = \arg\max_{j,s} G_m(j, s, t).$$

8. Stop if the best gain is not finite, is less than or equal to min_gain, or, when min_relative_gain is positive, does not exceed the required share of the parent-node impurity.
9. Otherwise split node t into child nodes t_L and t_R, then repeat the procedure recursively.

The same recursive logic is shown as a flowchart in Figure 14. The procedures are implemented in the ineqTrees package (Mburu, 2026). The fitted tree is stored using the partykit tree representation for plotting and traversal (Hothorn and Zeileis, 2015).

10.4.1 Flowchart of the Greedy Concentration-Index Tree-Growing Procedure

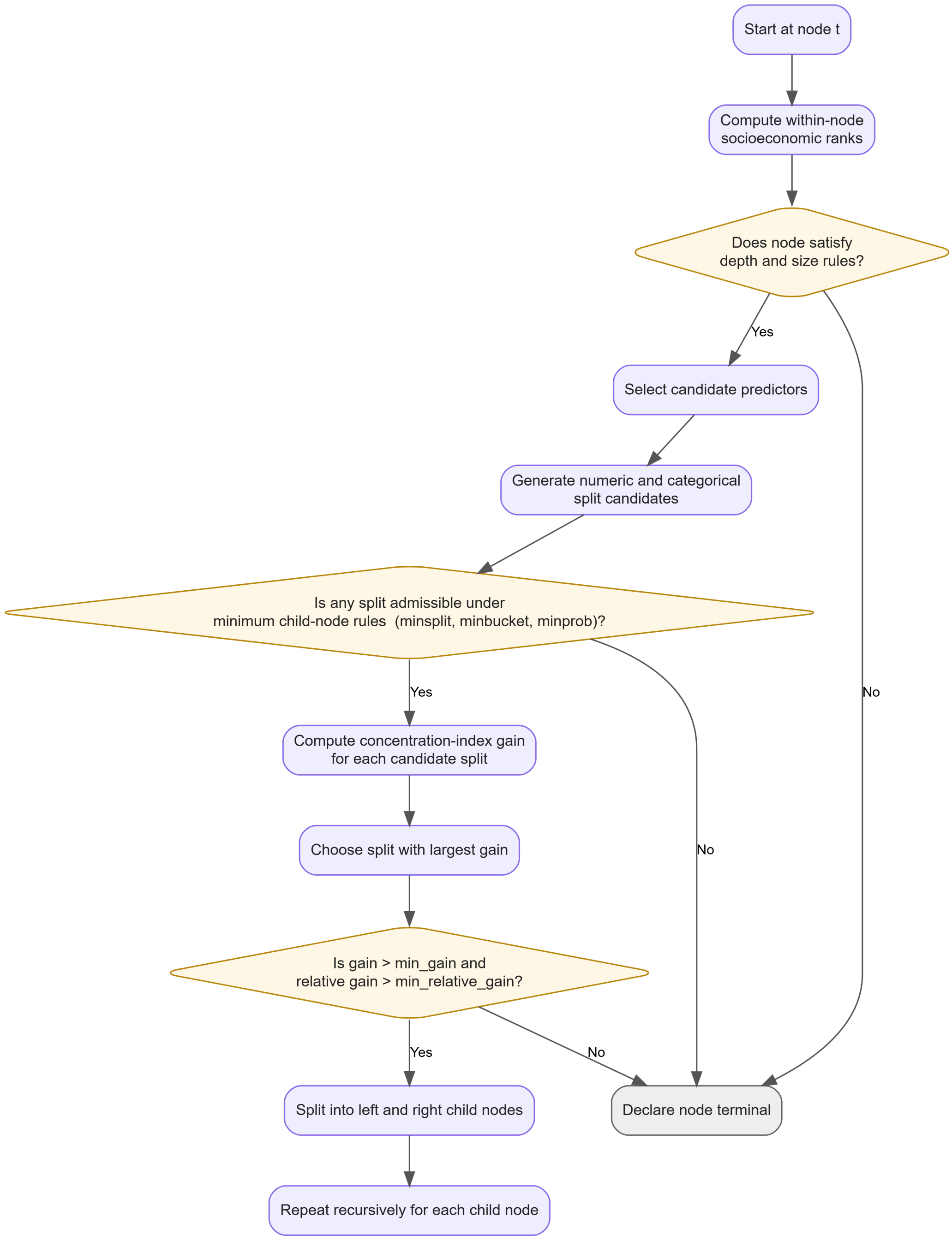

Figure 14: Greedy concentration-index tree-growing procedure.

10.5 Results

10.5.1 Appendix R1: Exploratory Data Outputs

Table 3: Weighted distribution of analysis variables by under-five mortality status

	Under-five mortality	Overall

Variable	Alive at age 5	Died before age 5	Overall
	(weighted N=7,464)	(weighted N=859)	(weighted N=8,323)
Household wealth			
High household wealth	4,266.9 (57.2%)	418.6 (48.7%)	4,685.5 (56.3%)
Low household wealth	3,197.4 (42.8%)	440.2 (51.3%)	3,637.6 (43.7%)
Skilled.birth.attendance			
Skilled.birth.attendance.1	3,225.3 (43.2%)	295.1 (34.4%)	3,520.5 (42.3%)
No.skilled.birth.attendance	4,239 (56.8%)	563.6 (65.6%)	4,802.6 (57.7%)
Sex of the child			
Female	3,828.8 (51.3%)	390.4 (45.5%)	4,219.2 (50.7%)
Male	3,635.5 (48.7%)	468.4 (54.5%)	4,103.9 (49.3%)
Birth order and interval			
First birth	1,396.8 (18.7%)	190.9 (22.2%)	1,587.7 (19.1%)
2-4 short interval	774.3 (10.4%)	110.1 (12.8%)	884.4 (10.6%)
2-4 long interval	2,603.8 (34.9%)	216.6 (25.2%)	2,820.3 (33.9%)
5+ short interval	696.6 (9.3%)	150.6 (17.5%)	847.2 (10.2%)
5+ long interval	1,992.8 (26.7%)	190.7 (22.2%)	2,183.5 (26.2%)
Mother's age at birth			
20 or more	4,256.1 (57.0%)	472.6 (55.0%)	4,728.8 (56.8%)
Less than 20	3,208.1 (43.0%)	386.2 (45.0%)	3,594.3 (43.2%)
Type of residence			
Urban	2,932.1 (39.3%)	263.6 (30.7%)	3,195.7 (38.4%)
Rural	4,532.2 (60.7%)	595.2 (69.3%)	5,127.4 (61.6%)
Mother's education			
Any education	3,342.9 (44.8%)	278.8 (32.5%)	3,621.8 (43.5%)
No education	4,121.4 (55.2%)	579.9 (67.5%)	4,701.3 (56.5%)
Father's education			
Any education	5,490 (73.6%)	589.1 (68.6%)	6,079.2 (73.0%)
No education	1,974.3 (26.4%)	269.7 (31.4%)	2,243.9 (27.0%)
Mother's occupation			
Other	1,610.2 (21.6%)	135.1 (15.7%)	1,745.4 (21.0%)
Household, unskilled, not working	1,757.2 (23.5%)	172.1 (20.0%)	1,929.4 (23.2%)
Agriculture	4,096.8 (54.9%)	551.6 (64.2%)	4,648.4 (55.8%)
Father's occupation			
Other	2,810.3 (37.6%)	242.1 (28.2%)	3,052.4 (36.7%)
Household, unskilled, not working	961.3 (12.9%)	108.5 (12.6%)	1,069.8 (12.9%)
Agriculture	3,692.7 (49.5%)	508.1 (59.2%)	4,200.9 (50.5%)

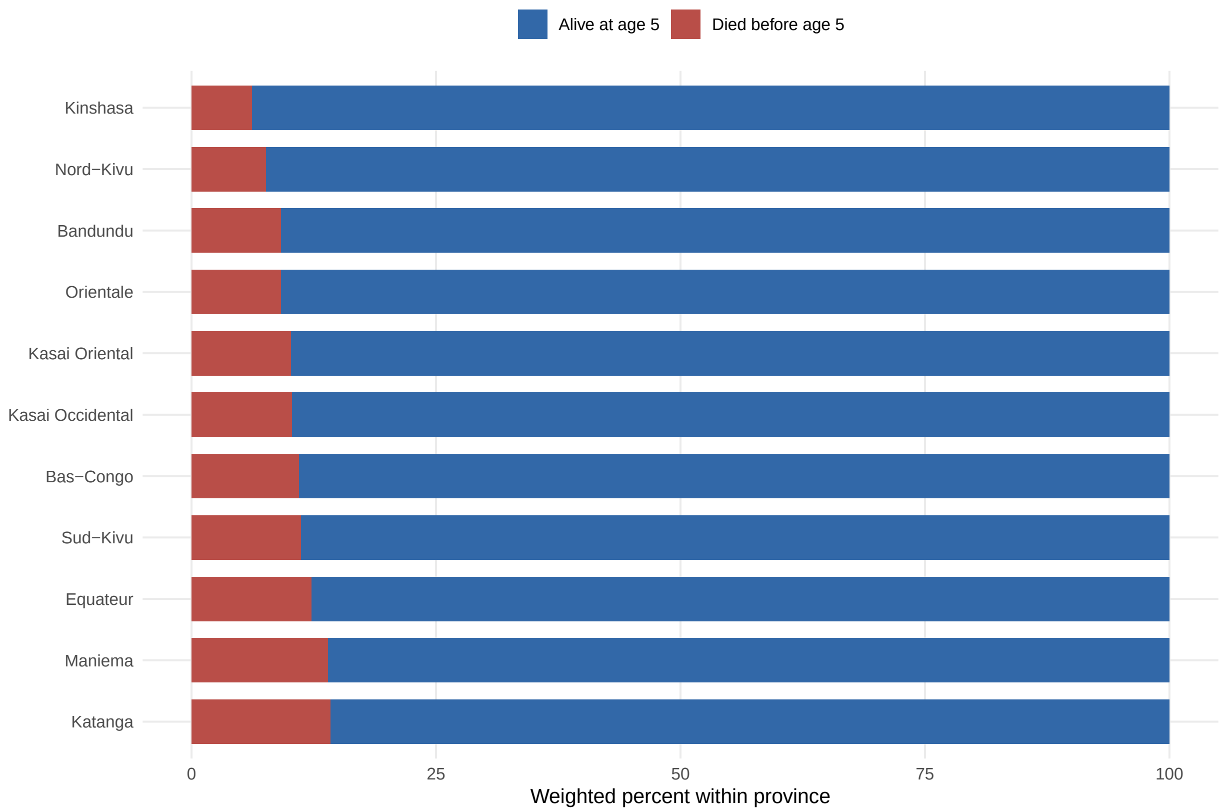

Figure 15: Weighted province distribution by under-five mortality status.

10.5.2 Appendix R2: Regression and Predictive-Forest Details

Regression model summary

Table 4: Survey-weighted regression model summary

Term	Estimate	Std. error	Statistic	p-value	Conf. low	Conf. high
Intercept	-2.908	0.282	-10.319	< 0.001	-3.463	-2.354
Household wealth	0.068	0.118	0.579	0.56311	-0.164	0.300
Skilled birth attendance	0.198	0.134	1.479	0.14018	-0.066	0.462
Sex of the child	0.242	0.084	2.864	0.00451	0.076	0.408
Birth order and interval: 2-4 short interval	-0.004	0.165	-0.023	0.98177	-0.328	0.320
Birth order and interval: 2-4 long interval	-0.509	0.136	-3.742	< 0.001	-0.777	-0.241
Birth order and interval: 5+ short interval	0.420	0.187	2.247	0.02543	0.052	0.787
Birth order and interval: 5+ long interval	-0.396	0.189	-2.096	0.03701	-0.769	-0.024
Mother's age at birth	0.033	0.152	0.215	0.82980	-0.267	0.332
Type of residence	0.090	0.135	0.667	0.50556	-0.176	0.356
Mother's education	0.336	0.120	2.799	0.00549	0.100	0.573
Father's education	-0.036	0.119	-0.300	0.76476	-0.270	0.199

Mother's occupation: Household, unskilled, not working	-0.018	0.159	-0.112	0.91123	-0.330	0.294
Mother's occupation: Agriculture	0.140	0.181	0.776	0.43864	-0.216	0.497
Father's occupation: Household, unskilled, not working	0.235	0.144	1.638	0.10253	-0.048	0.518
Father's occupation: Agriculture	0.225	0.143	1.577	0.11587	-0.056	0.506
Region: Bas-Congo	0.258	0.260	0.993	0.32139	-0.253	0.769
Region: Equateur	0.250	0.283	0.882	0.37862	-0.308	0.807
Region: Kasai Occidental	0.105	0.269	0.391	0.69603	-0.424	0.634
Region: Kasai Oriental	0.153	0.212	0.719	0.47266	-0.265	0.571
Region: Katanga	0.515	0.256	2.014	0.04499	0.012	1.018
Region: Kinshasa	0.081	0.288	0.283	0.77733	-0.485	0.648
Region: Maniema	0.489	0.235	2.084	0.03808	0.027	0.951
Region: Nord-Kivu	-0.252	0.319	-0.789	0.43053	-0.879	0.376
Region: Orientale	-0.084	0.295	-0.283	0.77728	-0.665	0.498
Region: Sud-Kivu	0.300	0.251	1.194	0.23338	-0.194	0.794

Predictive ranger tuning results

Table 5: Weighted class distribution before and after ranger undersampling

Sample	Outcome	Rows	Weighted N	Weighted percent
Before undersampling	Alive at age 5	7424	7464.30	89.68
Before undersampling	Died before age 5	872	858.80	10.32
After undersampling	Alive at age 5	1274	1289.74	60.03
After undersampling	Died before age 5	872	858.80	39.97

Table 6: Cross-validation results for the predictive ranger model

Metric	mtry	min_n	Mean	Std. error	Folds	Rank
accuracy	10	800	0.5896	0.0221	10	1
accuracy	22	200	0.5849	0.0166	10	2
accuracy	29	500	0.5844	0.0191	10	3
accuracy	16	800	0.5836	0.0203	10	4
accuracy	10	500	0.5820	0.0223	10	5
accuracy	22	800	0.5808	0.0161	10	6
accuracy	16	200	0.5806	0.0170	10	7
accuracy	22	500	0.5782	0.0209	10	8
accuracy	10	200	0.5781	0.0184	10	9

accuracy	29	200	0.5767	0.0153	10	10
accuracy	16	500	0.5765	0.0214	10	11
accuracy	29	800	0.5757	0.0158	10	12
f_meas	22	200	0.3776	0.0214	10	1
f_meas	29	200	0.3730	0.0209	10	2
f_meas	16	200	0.3627	0.0216	10	3
f_meas	10	200	0.3380	0.0207	10	4
f_meas	29	500	0.3358	0.0258	10	5
f_meas	22	500	0.3195	0.0300	10	6
f_meas	16	500	0.2945	0.0296	10	7
f_meas	10	500	0.2769	0.0305	10	8
f_meas	29	800	0.2498	0.0393	10	9
f_meas	22	800	0.2442	0.0374	10	10
f_meas	16	800	0.2315	0.0360	10	11
f_meas	10	800	0.1924	0.0348	10	12
roc_auc	10	500	0.6012	0.0114	10	1
roc_auc	16	800	0.6002	0.0111	10	2
roc_auc	16	200	0.5997	0.0093	10	3
roc_auc	16	500	0.5995	0.0101	10	4
roc_auc	29	800	0.5994	0.0094	10	5
roc_auc	10	800	0.5993	0.0120	10	6
roc_auc	22	500	0.5991	0.0094	10	7
roc_auc	22	200	0.5986	0.0099	10	8
roc_auc	22	800	0.5983	0.0104	10	9
roc_auc	29	200	0.5980	0.0092	10	10
roc_auc	10	200	0.5980	0.0103	10	11
roc_auc	29	500	0.5977	0.0088	10	12
sensitivity	22	200	0.3074	0.0296	10	1
sensitivity	29	200	0.3074	0.0287	10	2
sensitivity	16	200	0.2913	0.0284	10	3
sensitivity	10	200	0.2627	0.0269	10	4
sensitivity	29	500	0.2593	0.0322	10	5
sensitivity	22	500	0.2467	0.0345	10	6
sensitivity	16	500	0.2204	0.0330	10	7
sensitivity	29	800	0.1987	0.0543	10	8
sensitivity	10	500	0.1986	0.0307	10	9
sensitivity	22	800	0.1860	0.0477	10	10
sensitivity	16	800	0.1653	0.0384	10	11
sensitivity	10	800	0.1240	0.0280	10	12
spec	10	800	0.9457	0.0152	10	1
spec	16	800	0.9127	0.0227	10	2
spec	22	800	0.8967	0.0323	10	3
spec	29	800	0.8826	0.0364	10	4
spec	10	500	0.8783	0.0178	10	5
spec	16	500	0.8539	0.0218	10	6

spec	22	500	0.8367	0.0230	10	7
spec	29	500	0.8365	0.0224	10	8
spec	10	200	0.8227	0.0208	10	9
spec	16	200	0.8045	0.0217	10	10
spec	22	200	0.8005	0.0229	10	11
spec	29	200	0.7859	0.0216	10	12

Predictive ranger SHAP summary plot

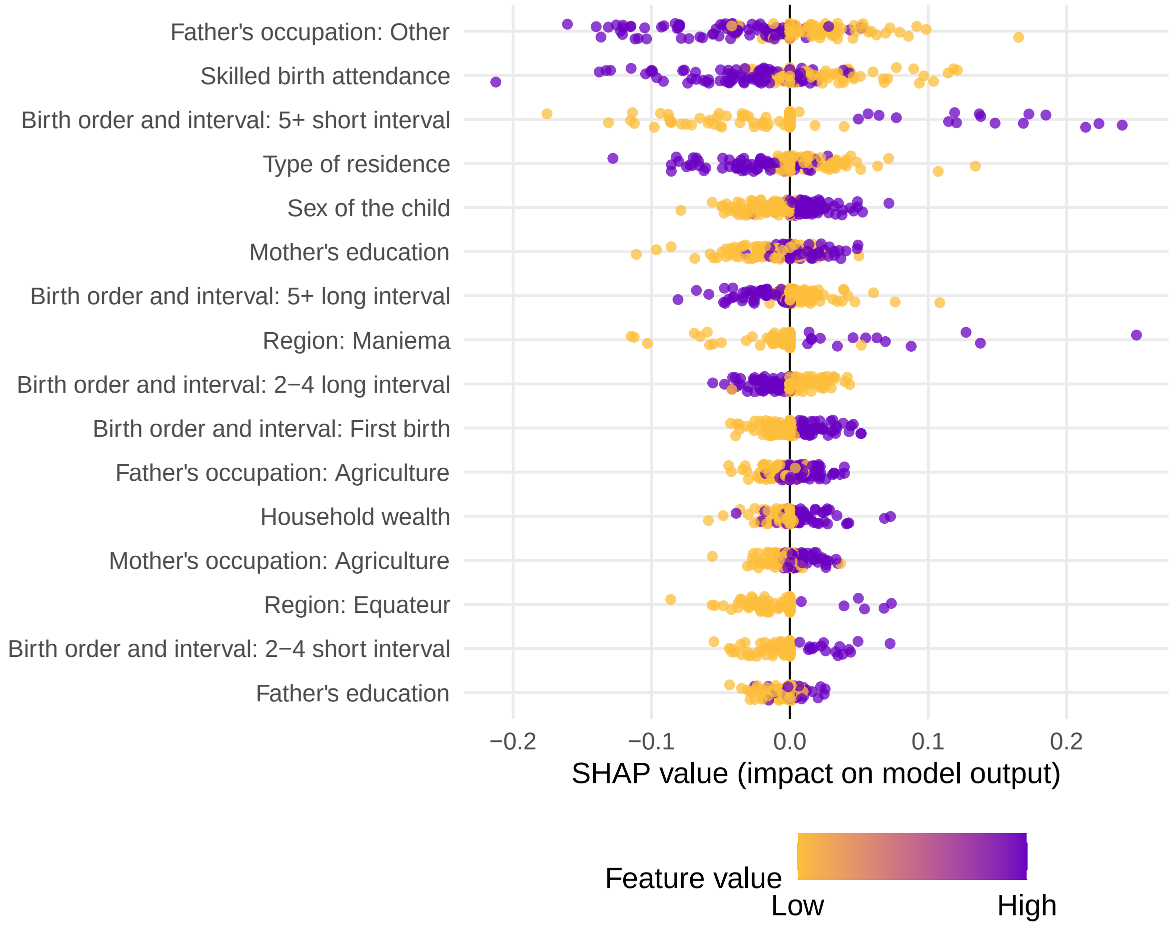

Figure 16: SHAP summary plot for the predictive ranger model.

10.5.3 Appendix R3: Concentration-Index Tree Details

Tree tuning and selection details

Table 7: Best tree hyperparameters selected by validation gain within each impurity criterion

type	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	500	100	0.01	10	1e-04	0.25
CIg	500	100	0.01	10	1e-04	0.30
CIc	500	100	0.01	10	1e-04	0.30

L	500	100	0.01	10	1e-04	0.25

Tree fit and selection summaries

Table 8: Tree summary for the selected settings

criterion	metric	estimate	std_error
CI	Root impurity	0.1293	NA
CI	Training gain	0.0960	0.0044
CI	Training gain (% root)	83.3135	1.6199
CI	Validation gain	-0.0234	0.0189
CI	Validation gain (% root)	-1157.1539	1131.8743
CIc	Root impurity	0.0591	NA
CIc	Training gain	0.0388	0.0019
CIc	Training gain (% root)	81.5691	2.1079
CIc	Validation gain	-0.0030	0.0077
CIc	Validation gain (% root)	-1058.0175	1039.5234
CIg	Root impurity	0.0148	NA
CIg	Training gain	0.0097	0.0005
CIg	Training gain (% root)	81.5691	2.1079
CIg	Validation gain	-0.0007	0.0019
CIg	Validation gain (% root)	-1058.0175	1039.5234
L	Root impurity	1.8028	NA
L	Training gain	0.1995	0.0174
L	Training gain (% root)	98.1872	0.3555
L	Validation gain	1.7776	1.6048
L	Validation gain (% root)	64.4907	11.3190

Tree selection metrics

Table 9: Selection metrics for fitted concentration-index trees

Criterion	Root objective	Train gain	Train gain/root	Validation gain	Validation gain/root	Terminal nodes
CI	0.1293	0.0960	0.8331	-0.0234	-11.5715	6.9
CIc	0.0591	0.0388	0.8157	-0.0030	-10.5802	6.0
CIg	0.0148	0.0097	0.8157	-0.0007	-10.5802	6.0
L	1.8028	0.1995	0.9819	1.7776	0.6449	7.5

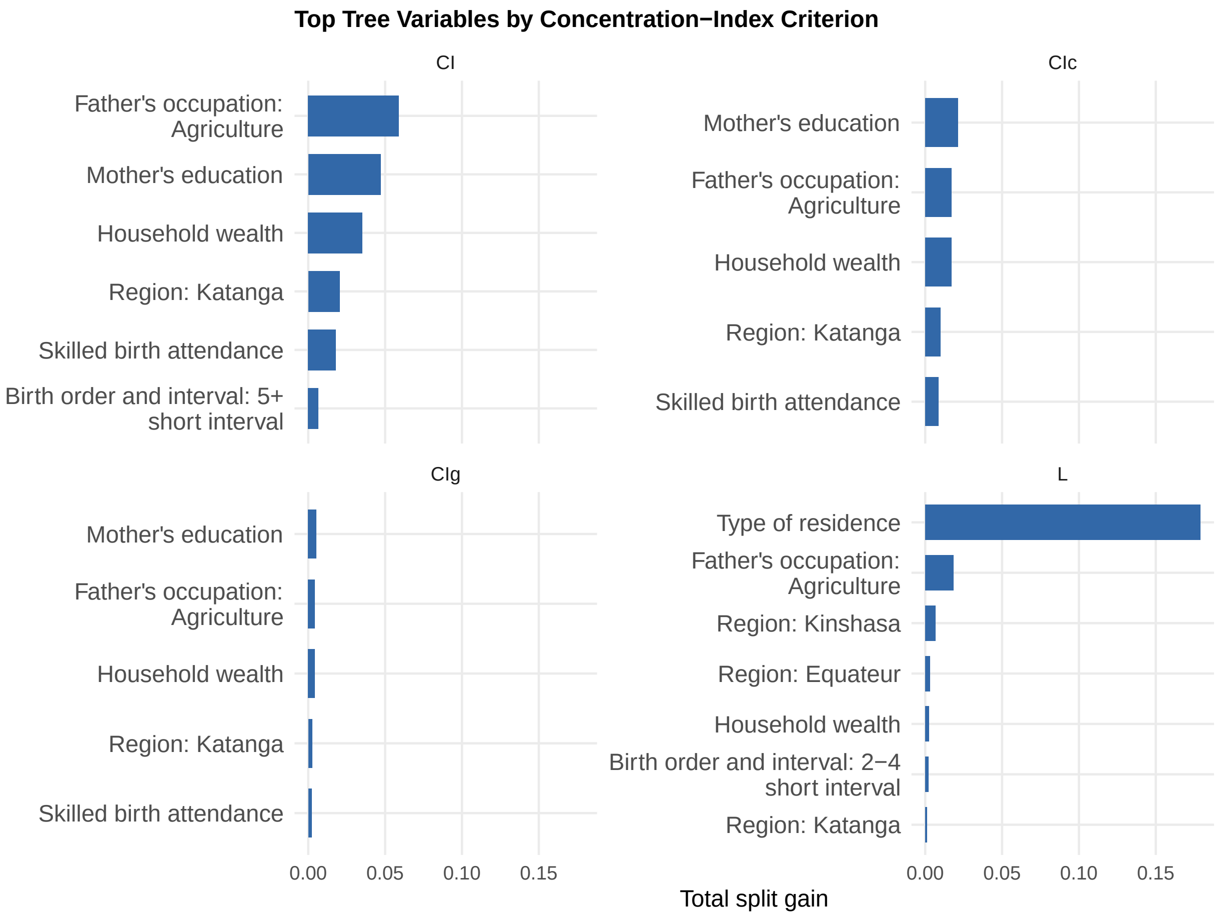

Figure 17: Top tree variables by concentration-index criterion.

10.5.4 Additional Concentration-Index Tree Plots

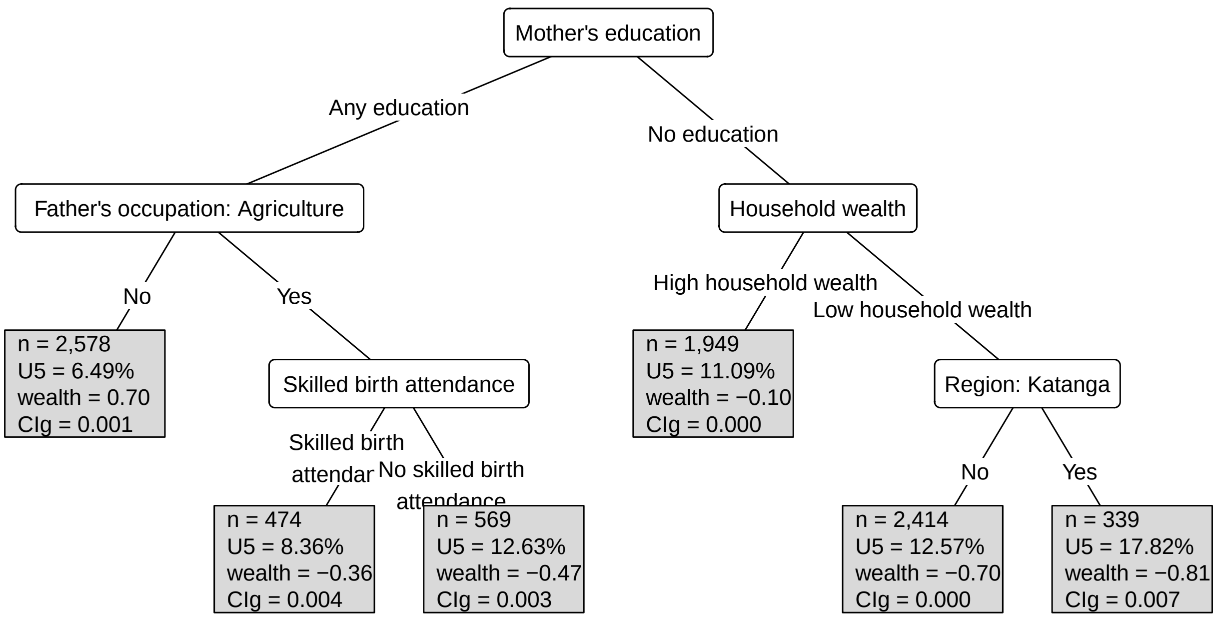

Figure 18: Concentration-index tree fitted with the CIg criterion.

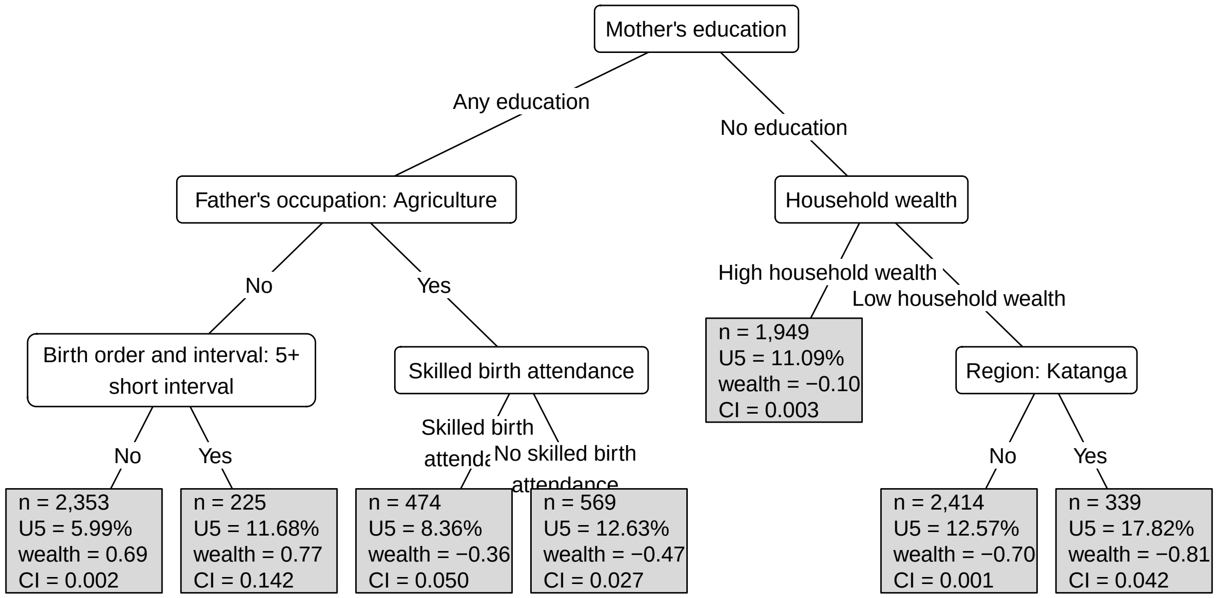

Figure 19: Concentration-index tree fitted with the CI criterion.

10.5.5 Appendix R4: Rpart Comparison Tree Details

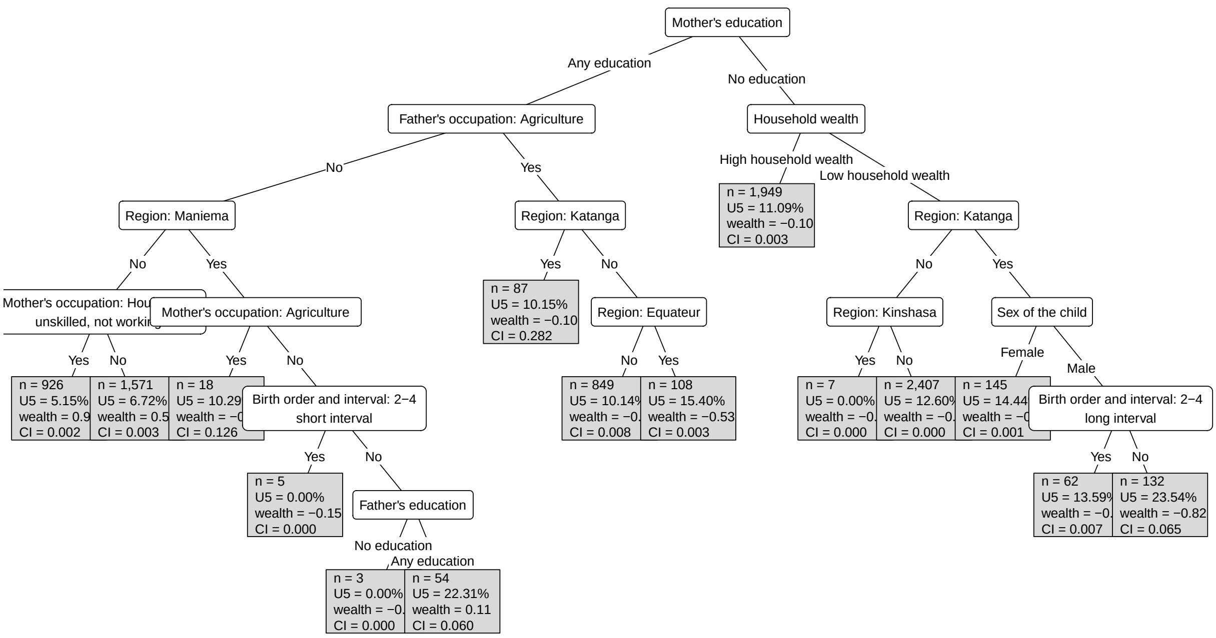

Figure 20: Rpart comparison tree using absolute CI impurity.

10.5.6 Appendix 5: Concentration-Index Forest Details

Forest tuning and selection details

Table 10: Best forest hyperparameters selected by validation gain within each impurity criterion

type	ntree	mtry	minsplit	minbucket	minprob	maxdepth	min gain	min relative gain

CI	100	15	500	100	0.01	15	0	0.25
CIc	100	15	500	100	0.01	15	0	0.25
CIg	100	15	500	100	0.01	15	0	0.25
L	50	15	500	100	0.01	15	0	0.20

Forest fit and selection summaries

Table 11: Forest summary for the selected settings

criterion	metric	estimate	std_error
CI	Root impurity	0.1293	NA
CI	Training gain	0.0553	0.0031
CI	Training gain (% root)	48.1137	1.9835
CI	Validation gain	-0.0137	0.0147
CI	Validation gain (% root)	-1038.2520	1024.0575
CIc	Root impurity	0.0591	NA
CIc	Training gain	0.0245	0.0011
CIc	Training gain (% root)	51.7127	1.8192
CIc	Validation gain	-0.0012	0.0075
CIc	Validation gain (% root)	-1108.4715	1101.3651
CIg	Root impurity	0.0148	NA
CIg	Training gain	0.0061	0.0003
CIg	Training gain (% root)	51.5535	1.9667
CIg	Validation gain	-0.0001	0.0019
CIg	Validation gain (% root)	-1063.0446	1055.1327
L	Root impurity	1.8028	NA
L	Training gain	0.1941	0.0173
L	Training gain (% root)	95.2953	0.6298
L	Validation gain	1.7544	1.5950
L	Validation gain (% root)	56.6798	11.5910

Forest selection metrics

Table 12: Selection metrics for fitted concentration-index forests

Criterion	Root objective	Train gain	Train gain/root	Validation gain	Validation gain/root	Mean terminal nodes
CI	0.1293	0.0553	0.4811	-0.0137	-10.3825	4.014
CIc	0.0591	0.0245	0.5171	-0.0012	-11.0847	4.113
CIg	0.0148	0.0061	0.5155	-0.0001	-10.6304	4.148
L	1.8028	0.1941	0.9530	1.7544	0.5668	5.448

Forest complexity plots

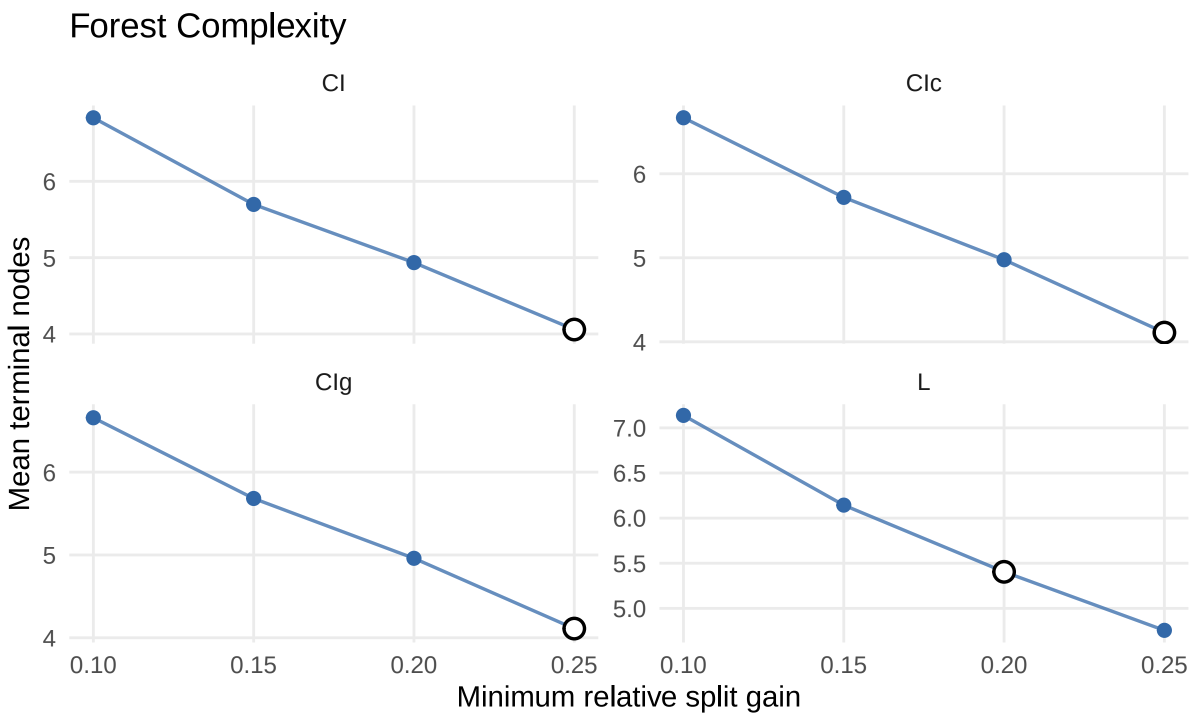

Figure 21: Forest complexity by number of terminal nodes.

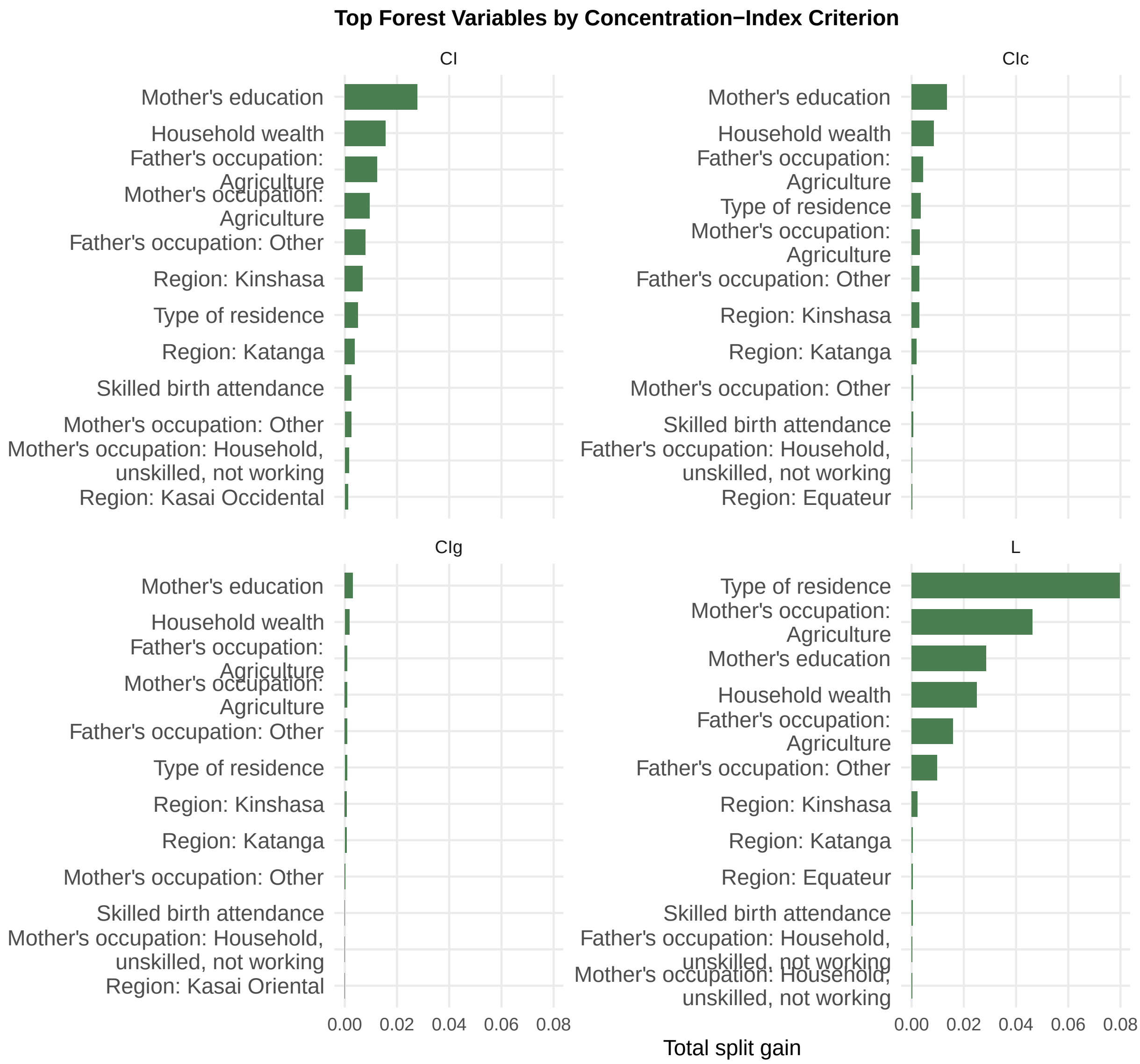

Figure 22: Top forest variables by concentration-index criterion.

Forest Surrogate Trees plots

The selected forest surrogate appears in the results section. The remaining criterion-specific forest surrogates are retained here for comparison.

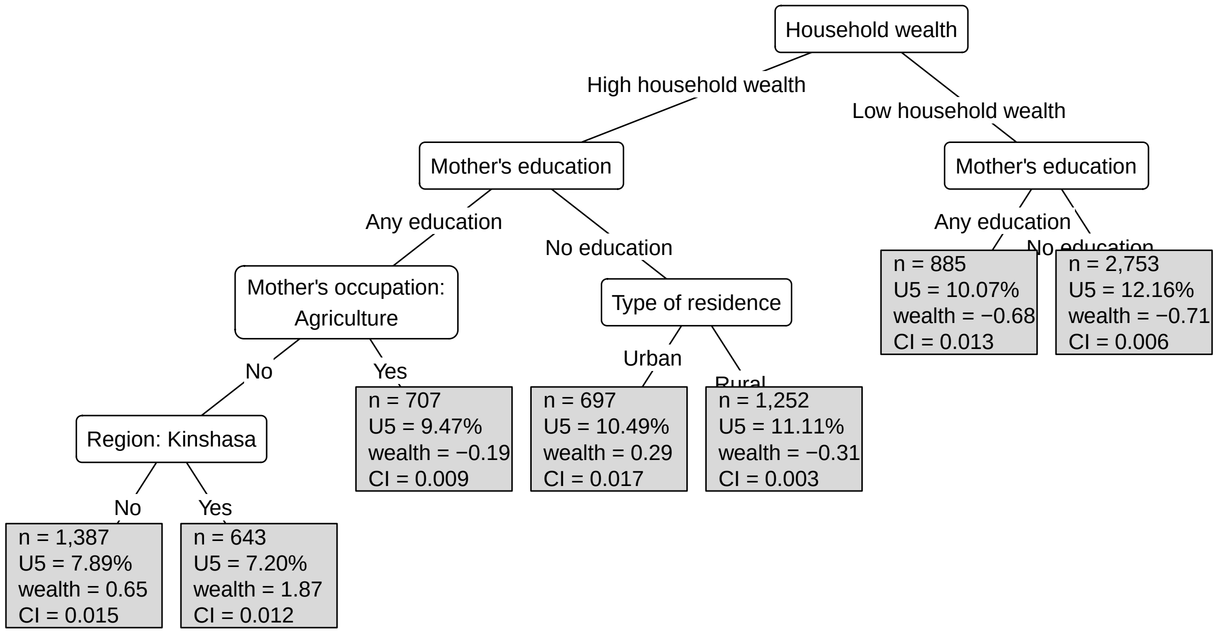

Figure 23: Surrogate tree for the CI concentration-index forest.

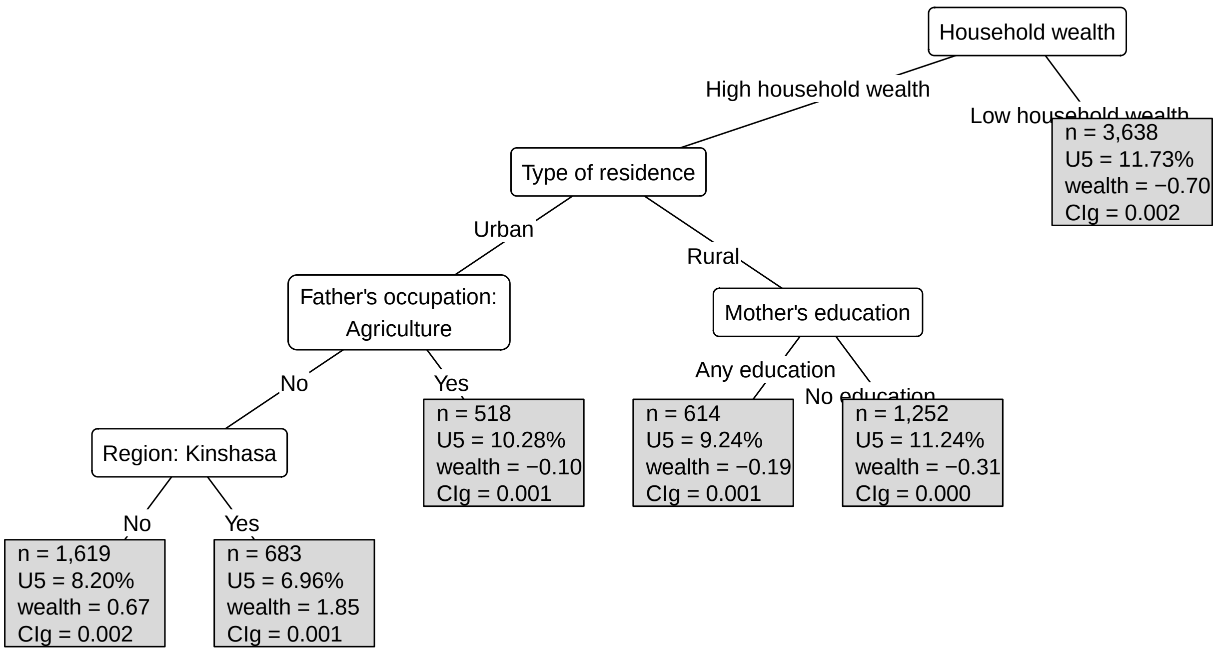

Figure 24: Surrogate tree for the CIg concentration-index forest.

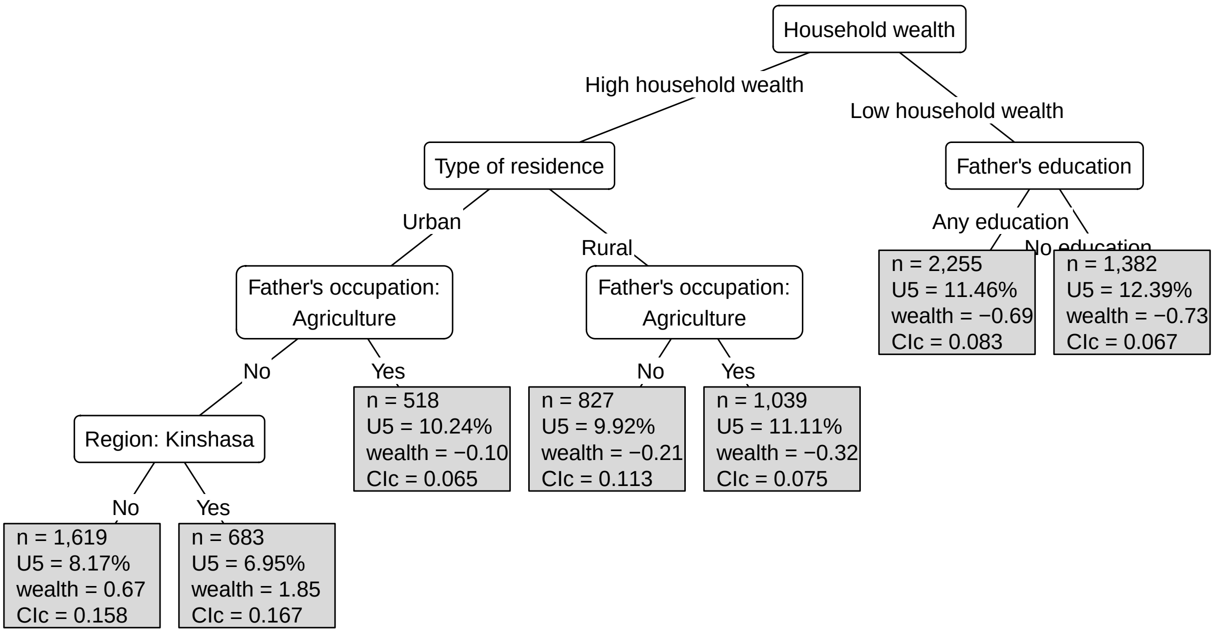

Figure 25: Surrogate tree for the CIc concentration-index forest.

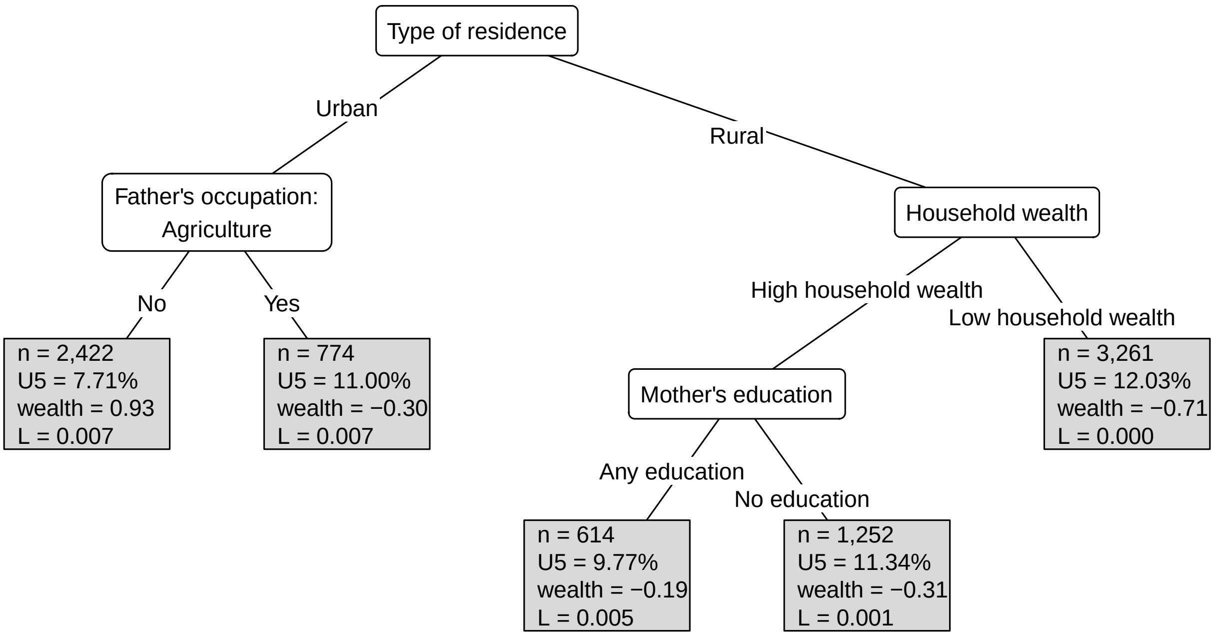

Figure 26: Surrogate tree for the L concentration-index forest.

10.5.7 Appendix S: Simulation Sensitivity Plots

The CIg tree is shown in the main results section. The remaining positive-control simulation trees are shown here to check whether the recovered subgroup pattern is stable across concentration-index criteria.

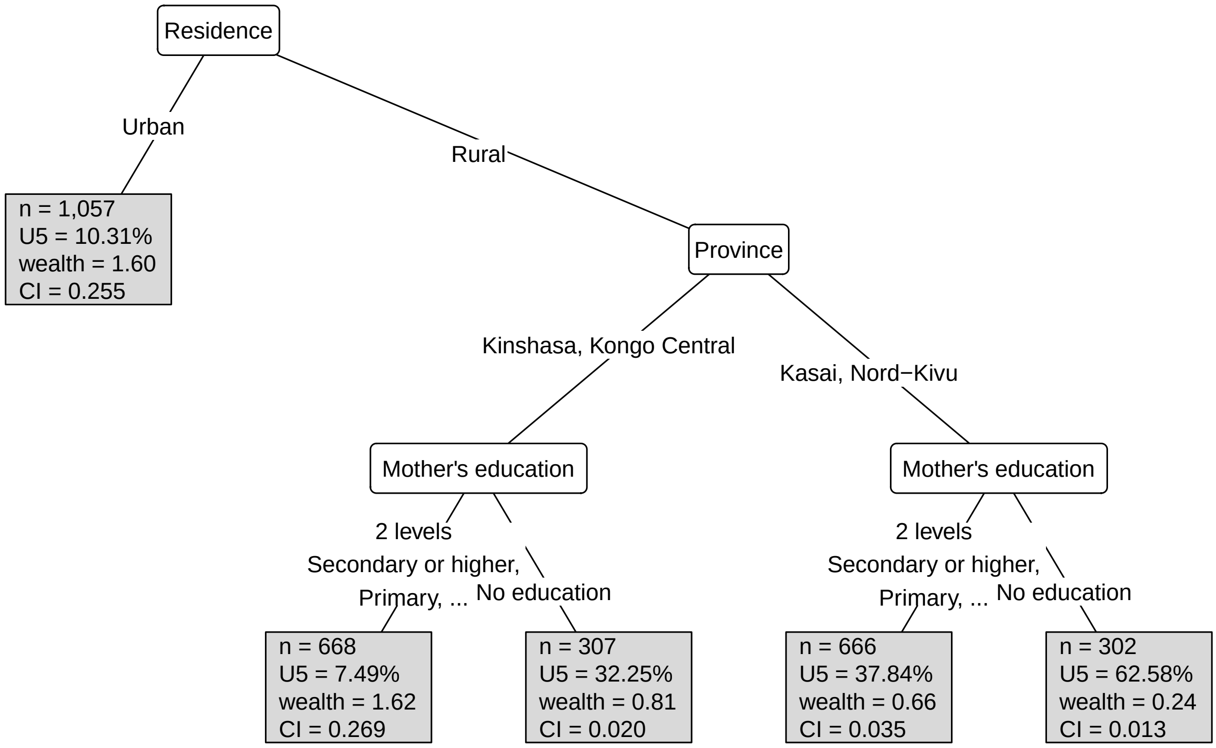

Figure 27: Positive-control simulation tree fitted with the CI criterion.

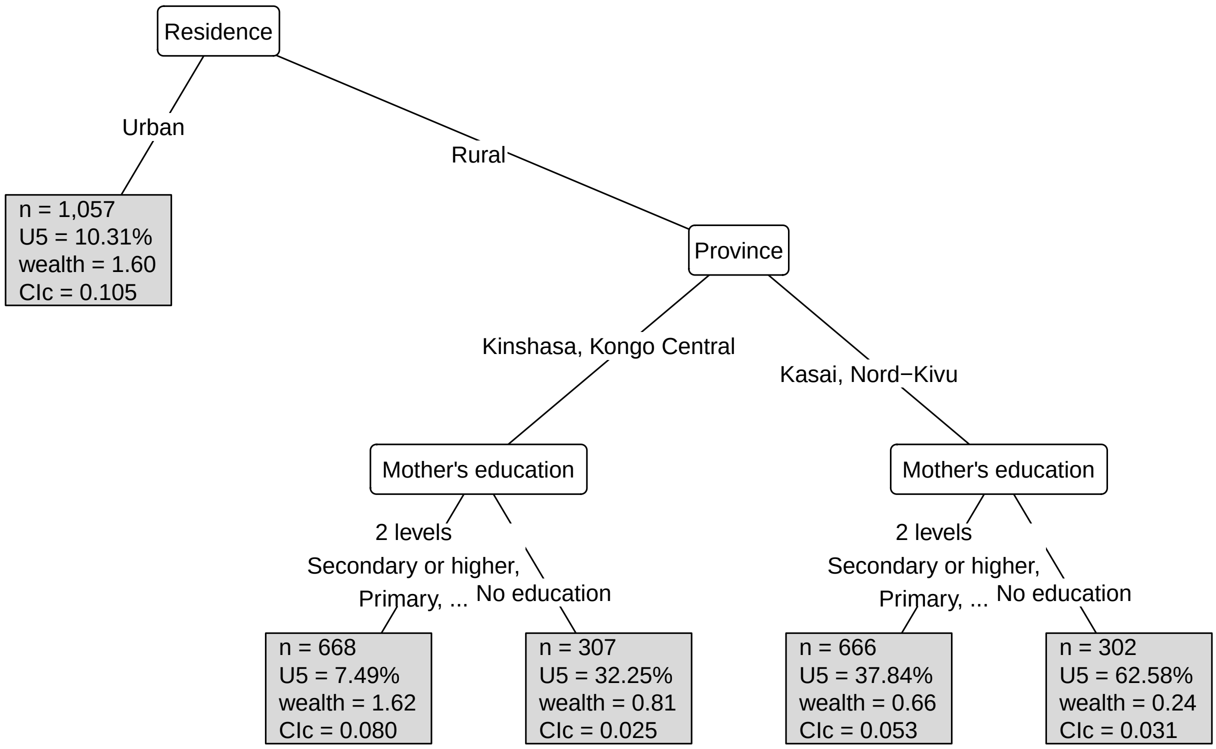

Figure 28: Positive-control simulation tree fitted with the CIc criterion.

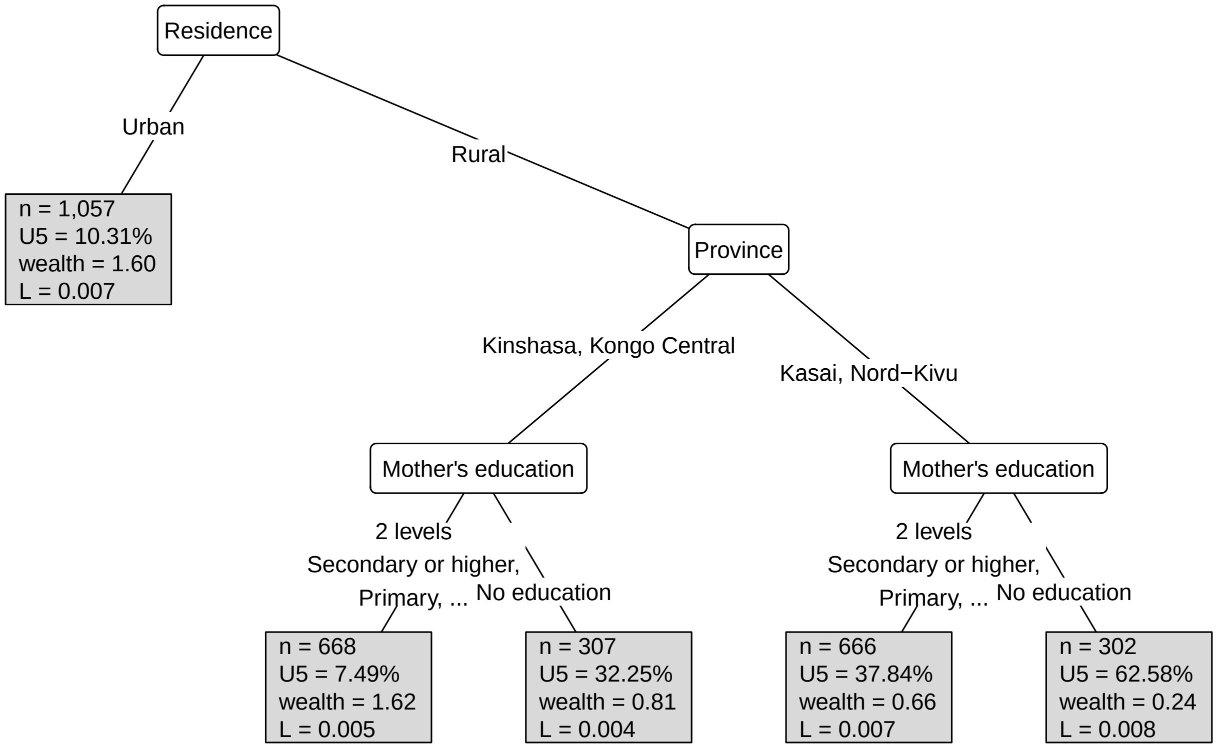

Figure 29: Positive-control simulation tree fitted with the L criterion.